

Embedded Spaces for Discovering New Classes

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Abstract

A representation can encode a novel class perfectly and still be useless for finding it. On a supervised contrastive encoder trained with one class held out, a linear probe separates that class from the rest at an AUC of 0.94, while the same space, asked to flag those examples without being told what to look for, flags none of them. Representation quality is routinely measured on the first side of this gap — linear probes, k -NN, transfer — while scientific discovery lives entirely on the second. We study the standard open-world loop (score a corpus against a labelled reference, pool the most anomalous points, cluster the pool into candidate class anchors, adapt the representation) and take as the primary metric the fraction of that pool which is genuinely novel, since every later stage is downstream of it. Our contributions are as follows. First, an algebra for the pool: purity is $\Pi = \varepsilon_s b/q$ with a ceiling $\min(1, b/q)$ that depends on the pooling threshold alone, which separates “the representation is bad” from “the threshold is bad” — questions the literature does not currently distinguish. Second, a label-free rule for choosing that threshold, derived from the same Neyman–Pearson density ratio that scores the corpus; it recovers the oracle threshold exactly on synthetic data, and where the scorer is too weak to support a choice it declines to pool and reports the refusal as an output. Third, evidence that ranking training objectives by discoverability is not the same as ranking them by probe accuracy: on a leakage-free hundred-class benchmark, adding a sliced-isotropy regulariser to a supervised contrastive objective ties the linear probe and supervised accuracy while raising novelty-detection AUC from 0.233 to 0.620 — and the same addition is neutral to harmful at ten classes, so we state the dissociation as a many-class phenomenon. Fourth, a 2×2 decomposition of representation construction against the discovery loop, which finds the two to be substitutes more often than complements. Fifth, a construction that requires no training at all: the pretrained backbone, pooled by the density ratio against seen-class centroids, out-discovers every fine-tuned objective on the datasets where discovery works, so whether a corpus is discoverable can be read off before any training is done. Throughout we separate the $\sim 1\%$ and $\sim 10\%$ novelty regimes, which invert several conclusions; we report single-holdout results as spreads over the choice of held-out class, because that choice moves the numbers eight times more than the training seed; we report where the pipeline fails as carefully as where it works; and we rebuild every CIFAR result from random initialisation, because a hub-pretrained backbone has already seen the class it is being asked to discover.

1 Introduction

A representation can encode a novel class perfectly and still be useless for finding it. This is not a hypothetical failure mode; it is the ordinary situation. Train a supervised contrastive encoder on a set of known classes, hold one class out, and a linear probe trained to separate the held-out class from the rest reaches an AUC of 0.94. Now ask the same space a different question: given a *single* image and no knowledge of what is being looked for, is this one novel? Run as a hypothesis test at a 5% false-positive rate, that space flags none of them — its detection rate on genuinely novel images is no better than the false-positive rate itself. The information is present, and linearly accessible. It is simply not accessible to anyone who does not already have the answer.

The gap between those two numbers is the subject of this paper. It is the gap between a representation being *decodable* and being *discoverable*, and it matters because scientific discovery lives entirely on the second side of it. A search for a new particle, a new galaxy morphology, or a new failure mode of a deployed model does not come with labels for the thing being searched for. What

it has is a large corpus, a smaller reference sample of what is already understood, and a question: is there something here that the reference does not explain?

Most of the representation-learning literature optimises for the first side. Linear probe accuracy, k -NN accuracy, and transfer benchmarks all measure how well a space supports a decision that someone else has already specified. They are the right metrics for that purpose. They are close to uninformative about whether a space will surface structure nobody has named yet, and — as the two numbers above indicate — they can be actively misleading, because the training objectives that maximise them are not the objectives that make distances mean anything in absolute terms.

What we do. We treat discovery as the target and work backwards. The pipeline we study is the standard open-world loop: score every point in a corpus against a reference of labelled examples, take the most anomalous ones as a *pool*, cluster the pool to propose new class anchors, and adapt the representation to the proposal. Every stage of that loop is downstream of one quantity — the fraction of the pool that is genuinely novel, which we call *pool purity*. Purity is the only forward-looking metric in the pipeline: every other measure describes a space you already have, whereas purity predicts whether the next step will improve that space or corrupt it. When purity is low the loop does not merely fail to help. It clusters background, plants anchors on that background, and then fine-tunes the representation to make the spurious structure sharper — so a bad pool leaves the representation worse than not running the loop at all.

Contributions.

1. **An algebra for the pool, and a ceiling.** Writing b for the novel base rate, q for the pool fraction, and ε_s for the fraction of novel points retained, pool purity is $\Pi = \varepsilon_s b / q$, so the enrichment Π / b is exactly ε_s / q and $\Pi \leq \min(1, b / q)$. The ceiling is a property of the cut alone. It says that a pool threshold inherited without derivation — as ours was — can cap achievable purity at a level no representation can rescue, and it separates “the space is bad” from “the cut is bad” for the first time.
2. **A label-free rule for the cut.** Because purity requires the novel labels, tuning the cut on purity is oracle tuning. We derive the novel-count estimator that the Neyman–Pearson critic already provides, and select the tightest cut that retains enough *estimated* novel points to remain clusterable. The rule uses only the corpus and the labelled reference. On synthetic data with an exact critic it recovers the oracle cut exactly; on real embeddings it raises round-1 pool purity from 0.59 to 0.89 on the strongest space where the scorer supports a choice, and refuses — reporting that it has refused — where it does not.
3. **A regime split, stated as a condition rather than a caveat.** Single-class novelty at a rate of $\sim 1\%$ and multi-class novelty at $\sim 10\%$ are different problems with different answers: which pooling statistic wins, whether stricter thresholds help or hurt, and what purity is attainable at all invert between them. We report them separately throughout.
4. **A reduced set of objectives, and the regime in which they differ.** From a design cube of ten contrastive objectives we identify the few that earn a place for discovery specifically, and show that the ranking by discovery is not the ranking by probe. On a from-scratch CIFAR-100 lineage, adding sliced-isotropy to a supervised contrastive objective is a tie on the linear probe (0.937 vs. 0.940) and a tie on supervised top-1 (0.597 vs. 0.608, a gap within run-to-run variation), while raising every detection column — Euclidean AUC 0.233 $\rightarrow$ 0.620, tied-Mahalanobis 0.279 $\rightarrow$ 0.602. Across four holdout draws the CIFAR-100 dissociation survives as a mean effect — tied Mahalanobis +0.10 with a draw spread of comparable size — so we state it as a property of the average, not of every draw. The same comparison on a from-scratch CIFAR-10 lineage goes the other way on *both* currencies — decodability and discoverability — (probe 0.884 vs. 0.905, Mahalanobis 0.454 vs. 0.552), and does so on every one of four holdout draws: the dissociation is a many-class phenomenon, and we state it as one. On the few-class lineage the objective that holds both currencies, on every draw, is the supervised distance New Physics Learning Machine (NPLM) with a class-wise marginal (probe 0.938 ± 0.018 over draws, top-1 tied with SupCon, per-event 0.21 against SupCon’s

0.01); it is also the one leakage-free space whose fine-tuning loop earns its keep, pooling at 0.43 ± 0.02 with the whole held-out class present and 0.12 ± 0.01 from a 2% injected sample — against SupCon’s 0.02 ± 0.03 and 0.003 ± 0.005 — while raising the probe from 0.94 to 0.97. The boundary of the shortlist is the corpus itself: an instance-level term over the unlabelled corpus (the generalised category discovery loss) becomes the best space we measure when the whole novel class is present unlabelled, monotonically in that fraction, and contributes nothing at the $\leq 5\%$ contamination the rest of this paper concerns (Table 2).

5. **Two constructions, and a decomposition of what they buy.** We study a residual decomposition — train a second encoder on what the anchors fail to explain, then concatenate — and frozen-space discovery, in which only the anchors move. The residual’s largest gain survives a width-matched control (a same-width non-residual head recovers +0.02 of a +0.15 probe gain and loses geometry), and on leakage-free CIFAR-10 it is a two-currency gain over its parent on every one of four holdout draws (probe $+0.027 \pm 0.010$ with top-1 within 0.006 of the parent everywhere; on CIFAR-100 the probe gain is a tie and the calibration gain is what survives). Frozen-space discovery with a density-ratio pool is the one construction in the campaign that is both backbone- and holdout-independent: round-1 purity clears 0.15 on 143 of 144 Galaxy10 draw $\times$ seed cells (95% CI [0.96, 1.00]; median 0.58, cell means 0.39–0.58, the single exception at 0.10), 0.26–0.55 on all eight leakage-free CIFAR-10 objectives, and 0.19 or better on every random draw of DTD, where the fine-tuning loop on the same parents has a worst draw between 0.000 and 0.210 depending on the backbone. It needs no training: the pretrained backbone alone, pooled by the density ratio, reaches 0.56–0.60 purity on Galaxy10 and 0.26 on DTD on all 30 draw cells, above every fine-tuned objective on both (Table 6). Because both constructions can be composed with the discovery loop, we report them in a 2×2 ablation (construction $\times$ discovery) rather than as a single delta. The composition is superadditive in 12 of 20 cells but concentrated: the effect that takes per-event detection from nominal to $6\times$ nominal lives in one cell, while the two largest gains elsewhere turn out to be main effects of the construction and of discovery separately (Section 5). We state this narrowly because the decomposition is what makes it checkable.
6. **A leakage-free lineage for the claims that need one.** Every CIFAR cell built from hub-pretrained weights inherits an encoder that already saw the held-out class, which makes it unusable as evidence for discovery. We rebuild the CIFAR battery from random initialisation and report the pretrained cells only as an ablation. Dataset ordering throughout is by distance from pretraining rather than alphabetically, so the reader can see which claims rest on a backbone that could have seen the answer.

What this paper is not. We do not claim that a regulariser which makes embedded distance behave like a log-likelihood thereby improves discovery; we tested that directly and it does not hold (Section 7). We do not claim the pipeline works everywhere. A substantial part of what follows is a characterisation of where it does not, and why — in several cases the obstruction is the novel-class rate rather than anything about the representation, which is a limit no encoder can move. Nor is this a generalised category discovery method, and we do not compare against that benchmark: those systems fine-tune on the unlabelled corpus and are scored on a K -way partition with K given, which is not the problem posed here. We do, however, score the generalised category discovery *loss* under our own protocol (Section 6): it yields the best space we measure when the whole novel class is present unlabelled, its gain is monotone in that fraction, and it contributes nothing at the $\leq 5\%$ contamination this paper is about.

2 Related work

Scoring novelty in a learned space. The out-of-distribution detection literature scores individual points against a training distribution — by softmax confidence (Hendrycks and Gimpel, 2017), by a tied-covariance Mahalanobis distance to class means (Lee et al., 2018), by an energy (Liu et al., 2020), or by k -nearest-neighbour distance in a normalised feature space (Sun et al., 2022) — and is evaluated by AUROC, a ranking measure. Contrastive training is known to sharpen these scores

(Winkens et al., 2020; Tack et al., 2020); Ruff et al. (2021) survey the shallow and deep variants. Our Euclidean and tied-Mahalanobis columns are these scores, and our per-event power is the same score read at a fixed false-positive rate. What that literature does not ask is what happens after the score: which fraction of the corpus to keep, how pure the kept set is, and whether the kept set supports the next round. Those are the questions the pool algebra of Section 4.3 is about, and the “95th percentile of the reference” cut whose ceiling that algebra exposes is the standard operating point in this literature.

Open-set recognition and category discovery. Open-set recognition (Scheirer et al., 2013; Bendale and Boulton, 2016) asks a classifier to reject inputs from unseen classes; Vaze et al. (2022b) show that closed-set accuracy largely predicts open-set performance, which is the decodability side of the gap this paper measures. Novel category discovery (Han et al., 2019, 2020; Fini et al., 2021) and its generalised form (Cao et al., 2022; Vaze et al., 2022a; Wen et al., 2023) go further and *cluster* the unseen classes, but under assumptions we do not make: the unlabelled set is dominated by the novel classes (typically half of it or more), the number of novel classes is given or estimated from the labelled ones, and the score is a Hungarian-matched clustering accuracy. The problem here is the low-contamination end of that spectrum — one novel class at 1–10% of the corpus — where a K -way partition of the corpus is not the deliverable and the purity of the candidate pool is. Table 2 places the two regimes on one axis by scoring the GCD representation loss under our protocol as a function of how much of the novel class the training corpus contains.

Two-sample tests and the density ratio. The Neyman–Pearson lemma (Neyman and Pearson, 1933) makes the likelihood ratio the optimal test statistic, and Wilks’ theorem (Wilks, 1938) gives its null distribution; the New Physics Learning Machine (D’Agnolo and Wulzer, 2019; D’Agnolo et al., 2021) fits that ratio with a neural network trained on the loss of Equation 2, and its kernel form (Letizia et al., 2022) and the sparse local-kernel ensemble of Grosso et al. (2025) are the scorers used here. Maximum mean discrepancy (Gretton et al., 2012) is the kernel two-sample test we report alongside them, and the classification- without-labels construction (Metodiev et al., 2017) and the LHC Olympics (Kasieczka et al., 2021) are the physics setting in which signal-agnostic searches of this kind are benchmarked. All of these are dataset-level tests: they return one p -value for the corpus. Our use of the same statistic to score, pool and count individual points, and the identification of the pooled count with a mixture-proportion estimate, is the step from a test to a discovery loop. That estimate is the semi-supervised novelty-detection model of Blanchard et al. (2010) — corpus = $(1 - b)$ reference + b novel — whose class prior b is the mixture proportion that Elkan and Noto (2008) and Ramaswamy et al. (2016) estimate from the density ratio; the rank-matched null correction of Section 4.3 is what makes that estimate usable when the fitted ratio has spread of its own. Sugiyama et al. (2012) review density-ratio estimation more broadly.

Representation objectives and how they are evaluated. The objectives we compare are SimCLR (Chen et al., 2020), supervised contrastive learning (Khosla et al., 2020), the density-ratio objectives above used as contrastive losses, and the sketched isotropic-Gaussian regulariser of LeJEPa (Balestriero and LeCun, 2025), whose successor VISReg (Wu et al., 2026) supplies one of our pre-trained backbones alongside DINO (Caron et al., 2021). Balestriero and LeCun (2025) prove that an isotropic Gaussian embedding minimises worst-case downstream risk over linear probes; whether that guarantee transfers to a task that has no labels for the thing being sought is one of the questions this paper tests, and Section 7 reports that the mechanism one would expect — distance becoming a log-likelihood — is not what carries the effect. Representation quality is conventionally read from linear probes (Alain and Bengio, 2017), k -NN accuracy and transfer; our decodability columns are those measures, and the local-intrinsic-dimension estimator of Levina and Bickel (2005) is the geometry diagnostic we carry alongside them.

3 Setup: discovery as a two-sample problem

The open-world protocol. We are given a corpus $\mathcal{D}$ of N embedded points, of which an unknown fraction b belong to classes never seen in training, and a reference $\mathcal{R} \subset \mathcal{D}$ of points carrying known class labels. No label for any novel class is available at any point — not for choosing hyperpa-

rameters, not for setting thresholds. We state this explicitly because the constraint is easy to violate accidentally, and because a method that quietly violates it reports numbers that cannot be reproduced in the situation it claims to address.

The loop. One round proceeds as follows. Score every point by an anomaly score $s(x)$; threshold to obtain a pool $\mathcal{P}$; cluster $\mathcal{P}$ and append the resulting centroids to the anchor set; pseudo-label every pooled point by its nearest discovered anchor; and adapt the representation (or, in the frozen variant of Section 5, only the anchors) to the enlarged anchor set. The round repeats with the updated anchors.

Vocabulary. Six terms recur throughout. An *objective* (in tables, an *arm*) is one training loss from the design cube of Section 4. A *backbone* is the pretrained encoder an objective is fine-tuned on top of (DINO, LeJEPa or ViSReg; Caron et al., 2021; Balestriero and LeCun, 2025; Wu et al., 2026), or the from-scratch network where no pretraining is used. A *draw* is a choice of which class to hold out and treat as novel; re-drawing changes the problem, since some classes are far harder to find than others. A *seed* is a re-run of the same training with a different random initialisation and batch order, holding the draw fixed. A *cell* is one (dataset, backbone) pair, inside which objectives are compared. A *lineage* is a family of encoders sharing a pretraining origin: *leakage-free* encoders are trained from random initialisation, while *hub-pretrained* encoders use public pretrained weights and may have seen the held-out class. We hold out either a single class (the *single-holdout* regime, $b \approx 0.01$ – 0.1) or ten classes at once (the *multi-holdout* regime, $b \approx 0.2$), and report the two separately because several conclusions invert between them.

What we measure, and what the numbers mean. Because this paper compares two kinds of quality that are usually not distinguished, it is worth stating plainly what each number is before any of them appear.

Decodability measures how well a space supports a decision somebody has already specified. We report **linear probe AUC** (Alain and Bengio, 2017) (train a linear classifier to separate the held-out class from the rest, *using the held-out labels*, and measure how well it does) and **top-1 accuracy** on the seen classes. A probe is an oracle measurement: it is allowed the answer. It says whether the information about the novel class survives in the representation at all, which is a genuine and useful thing to know — and is not what a discovery pipeline can use.

Discoverability measures what the space supports *without* the answer. **Euclidean AUC** ranks every corpus point by its distance to the nearest known-class anchor and asks how well that ranking separates novel from seen; **tied Mahalanobis** does the same with a shared covariance rather than raw distance. Both are ranking measures: they say a novel point tends to score higher than a seen one, but not by how much or where to put a threshold. **Per-event power** is the strictest question we ask. Fix a decision threshold that admits $\alpha = 5\%$ of *reference* points, then report the fraction of genuinely novel points that exceed it. This is the power of a hypothesis test at size α , applied one event at a time. It is bounded below by α in the sense that a space carrying no usable signal scores $\approx \alpha$; a value at or under 0.05 therefore means the space cannot flag an individual novel example at all, and the interesting question is whether a method crosses that floor rather than by how much it improves. Finally, **pool purity** Π is defined below and is the metric this paper treats as primary. Alongside these per-event quantities we report *dataset-level* two-sample power — the probability that a kernel-based test (SparKer, Mahalanobis or maximum mean discrepancy; Grosso et al., 2025; Lee et al., 2018; Gretton et al., 2012) rejects the background-only hypothesis when a fraction f of novel examples is injected into a background sample — mainly in the appendix grids. We refer to this fixed suite of measurements as the *battery*.

Pool purity, and why everything is downstream of it. Let Π denote the fraction of $\mathcal{P}$ that is truly novel. The pool feeds directly into a clustering step with no supervision downstream of it, so $\mathcal{P}$ is the last point at which a mistake can still be corrected and nothing after it checks the work. At $\Pi \approx 0.01$ the clustering is not finding the novel class; it is partitioning background, and the subsequent fine-tune sharpens that partition. Empirically the probe drops by 0.031 to 0.038 wherever such pools form. We therefore report Π as the primary metric and compare spaces and procedures by it directly. Post-discovery quantities are reported in two regimes: the *whole-class pass*, in which the entire

held-out class is present unlabelled in the corpus the loop draws from, and the *injected pass*, in which discovery starts from a small injected sample of the held-out class (2% of the corpus unless stated). The injected pass is the paper’s operating regime; the whole-class pass is easier and is shown for reference. We deliberately do *not* impose a purity threshold: purity is monotone in the cut (Section 4.3), so any fixed level is a statement about the operating point rather than about the space, and no experiment in this paper branches on one. Where a purity is low enough that the subsequent clustering is partitioning background, we say so and give the number.

4 Algorithms

This section describes the objectives that produce the embedding, the statistic that scores it, and the rule that turns that statistic into a pool. The three are usually treated as separate literatures; a contribution of this work is that the middle one — the Neyman–Pearson density ratio — is the same object in all three roles, which is what makes the last one derivable rather than tuned.

4.1 Objectives

We work within a design cube spanning $\{\text{instance, supervised}\} \times \{\text{cosine, bilinear, distance}\} \times \{\text{softmax, NPLM}\} \times \{\text{none, SIGReg, class-wise SIGReg}\}$ marginals, which contains SimCLR, SupCon, LeJEPa/VISReg and the density-ratio objectives as corners (Chen et al., 2020; Khosla et al., 2020; Balestrieri and LeCun, 2025; Wu et al., 2026; D’Agnolo and Wulzer, 2019). Of the ten arms we ran, three earn a place for discovery.

SupCon + SIGReg (strong). Supervised positives, cosine critic, softmax estimator, and SIGReg, the sketched isotropic-Gaussian regulariser of LeJEPa (Balestrieri and LeCun, 2025), at weight $\lambda = 5$. This is the default discovery objective throughout: it produces the highest pool purities we measure. Two points of care. First, λ matters and is not a formality — the same objective at the library default $\lambda = 1$ is a different algorithm and its numbers must not be pooled with these (Appendix E). Second, we make no claim that SIGReg helps *because* it renders distance a log-likelihood; that mechanism is tested and rejected in Section 7. SIGReg earns its place empirically, and structurally: under an isotropic marginal Euclidean distance is Mahalanobis distance, which is what licenses the unit-variance clustering criterion used downstream.

Supervised distance-NPLM. Supervised positives, distance critic, and the estimator of the New Physics Learning Machine (NPLM; D’Agnolo and Wulzer, 2019),

$$\mathcal{L}[g] = \mathbb{E}_{\mathcal{R}}[e^g - 1] - \mathbb{E}_+[g], \quad (1)$$

whose minimiser is the *absolute* log density ratio, fixed by the calibration $\mathbb{E}_{\mathcal{R}}[e^{g^*}] = 1$ with no free additive constant. Unlike a softmax-normalised critic, this makes a single embedded distance a statement about relative likelihood rather than a statement only about rank. It is the strongest arm in the low-class-count regime and the most robust across pretraining backbones. Its class-wise-marginal variant (NPLM-dist. + classwise SIGReg) is the arm that holds both currencies on the leakage-free CIFAR-10 lineage and the supervised arm that the discovery loop improves most on the leakage-free CIFAR-100 lineage (+0.037 probe; Appendix B).

SupCon (plain). Supervised positives, cosine critic, softmax estimator, no marginal. Retained as the baseline and as the parent for the residual construction. It is the best space in the paper by linear probe and by semantic agreement, and one of the weakest by every calibration measure — which is the dissociation this paper is organised around, in a single arm.

We drop plain SimCLR, LeJEPa/SIGReg-SSL, instance NPLM-bilinear and instance NPLM-distance to a single ablation row: none reaches the Pareto front for discovery, and the bilinear critic in particular carries roughly three times the seed variance of the distance critic.

4.2 The Neyman–Pearson statistic, in three roles

Let $\mathcal{R}$ be the reference and $\mathcal{D}$ the corpus, with $w = N_{\mathcal{D}}/N_{\mathcal{R}}$. The empirical NP objective

$$\mathcal{L}[f] = w \sum_{x \in \mathcal{R}} (e^{f(x)} - 1) - \sum_{x \in \mathcal{D}} f(x) \quad (2)$$

has minimiser $f^*(x) = \log(p_{\mathcal{D}}(x)/p_{\mathcal{R}}(x))$, obtained by setting the functional derivative $p_{\mathcal{R}}e^f - p_{\mathcal{D}}$ to zero, and therefore satisfies

$$\mathbb{E}_{\mathcal{R}}[e^{f^*}] = \int p_{\mathcal{R}} \cdot \frac{p_{\mathcal{D}}}{p_{\mathcal{R}}} = 1. \quad (3)$$

Equation 2 is Equation 1 with pairs replaced by events. The same functional appears as a contrastive *loss* over pairs, as a two-sample *test statistic* $t_{NP} = -2\mathcal{L}$ distributed as a Wilks log-likelihood ratio (Wilks, 1938; D’Agnolo and Wulzer, 2019), and — below — as a *pool scorer*. We parameterise f as a small kernel ensemble with learnable centres and amplitudes (SparKer; Grosso et al., 2025) and anneal the bandwidth from the median pairwise distance downwards; a fixed bandwidth is calibrated only for one embedding scale and becomes uninformative when the geometry is rescaled.

Equation 3 is also a free diagnostic, and we report it. A fitted critic need not satisfy it, and a value far from 1 means the numbers built from that fit are unreliable. Measured on trained spaces it is 0.997 to 1.019 in-sample; out-of-sample it ranges to 2.7, reflecting mild overfitting of the reference subsample.

Why the ratio and not the distance. Distance-based pooling asks which points are far from everything. The density ratio asks which points are over-represented relative to what the reference explains. These come apart, and the difference is decisive: a novel class is not a scatter of far-flung outliers but a *dense clump* sitting where the reference has no mass. On a 100-class benchmark the extreme distance tail is dominated by background outliers, and tightening the distance threshold makes purity *worse* ($0.036 \rightarrow 0.002$), whereas tightening the ratio threshold makes it better ($0.214 \rightarrow 0.268$ with the encoder exactly frozen; an earlier archive that let BatchNorm statistics drift read $0.232 \rightarrow 0.358$, and we do not use it). Same space, same rate, opposite response.

4.3 Choosing the pool: an algebra and a rule

The algebra. Let $q = |\mathcal{P}|/N$ and let $\varepsilon_s, \varepsilon_b$ be the fractions of novel and background points retained. Then $q = \varepsilon_s b + \varepsilon_b(1 - b)$ and

$$\Pi = \frac{\varepsilon_s b}{q}, \quad E \equiv \frac{\Pi}{b} = \frac{\varepsilon_s}{q}, \quad \Pi \leq \min\left(1, \frac{b}{q}\right). \quad (4)$$

Three consequences. The enrichment is exactly the lift at operating point q . Purity increases monotonically as q shrinks, so there is no interior optimum and targeting a purity *level* is the wrong objective. And the ceiling $\min(1, b/q)$ is a property of the cut alone: at $q = 0.05$ and $b = 0.01$ no scorer, however good, can exceed $\Pi = 0.20$. A threshold chosen without reference to b therefore caps purity independently of the representation — and the widely used “95th percentile of the reference” is such a threshold.

The rule. Purity cannot be used to choose q , since computing it requires the novel labels. The critic supplies the missing quantity. Under the contamination model $p_{\mathcal{D}} = (1 - b)p_{\mathcal{R}} + b p_S$, with $r = e^f$, the posterior that a corpus point is novel is $1 - (1 - b)/r$; dropping the unknown b and taking the positive part gives the label-free weight

$$w(x) = \left[1 - \frac{1}{r(x)}\right]_+, \quad \mathbb{E}_{\mathcal{D}}[w] = \text{TV}(p_{\mathcal{D}}, p_{\mathcal{R}}) = b \text{TV}(p_S, p_{\mathcal{R}}) \leq b, \quad (5)$$

so $\sum_{x \in S} w(x)$ estimates how many novel points a subset S contains, exactly when the novel class is disjoint from the reference and biased *downward* under overlap — a safe direction, since it tightens the cut. Since purity rises as q falls, the binding constraint is not purity but whether a cluster survives, and that is what we target: ordering the corpus by f and writing $\hat{N}(k)$ for the null-corrected sum of w over the top k points,

$$k^* = \min \{ k : \hat{N}(k) \geq n_{\min} \}, \quad q^* = k^*/N, \quad k_{\max} = \left\lfloor \hat{N}(N) / n_{\min} \right\rfloor. \quad (6)$$

The same weights bound the clustering search range, replacing a bound that in the original pipeline was set from the number of held-out classes — a quantity an open-world method does not have.

The null correction, which is not optional. Equation 5 is a positive part, so *any* spread in a fitted f manufactures apparent novelty. On pure-noise scores with no novel class at all, the raw sum returns 665 “novel” points out of 8000 — enough for the rule to fire on nothing and hand the loop a pool of pure background. Subtracting the mean weight of the reference does not repair this, because top-ranked null points carry far above-average weight and their tail survives subtraction. We instead subtract a *rank-matched* null: under the null hypothesis the top q of the corpus is distributed as the top q of the reference, so $\hat{N}(k) = \sum_{i \leq k} w(x_{(i)}) - k \cdot \bar{w}_{\mathcal{R}}^{(q)}$, with $\bar{w}_{\mathcal{R}}^{(q)}$ the mean weight of the top- q fraction of the reference. This cancels to zero under the null while preserving a real excess. When $\hat{N}(N) < n_{\min}$ the rule declines to pool and reports that it has declined; that refusal is an output of the method, and we report it as one.

Constants. $n_{\min}$ and the clipping range on q are set once, globally, from a sweep on held-out embeddings rather than inherited: purity falls monotonically as $n_{\min}$ grows, so it is set as small as the clustering will bear.

5 Constructions

Both constructions in this section are defined *relative to an anchor set*, which is what lets them compose with the discovery loop: discovery enlarges and corrects the anchors, so a construction defined against them is a different construction after discovery than before. That composition is the part of this work with no counterpart we are aware of, and it is also the part most easily overstated, so we measure it as an interaction rather than a delta.

5.1 The residual decomposition

Let μ_y be the anchor for class y and z an embedding. The residual $r = z - \mu_y$ is what the anchor set fails to explain about z . We train a second encoder on r under the same objective family as the parent and either use it alone (*residual*) or concatenate it with the parent (*concat*).

The motivation is that a supervised objective is free to discard any variation that does not separate the classes it was given, and novelty is precisely variation of that kind. The residual target makes that discarded structure the only thing the child can see.

On its own, the residual adds detection at no accuracy cost. On the from-scratch CIFAR-100 lineage, concatenating a residual onto a supervised contrastive parent moves Euclidean novelty AUC from 0.233 to 0.521 while the linear probe moves by +0.0015, a gain two orders of magnitude smaller than the detection gain and of the same order as the probe’s own seed spread ($\sigma = 0.0008$). This is the pattern across cells: the pure residual usually *loses* probe accuracy (all four strong-parent cells, -0.037 to -0.077), and the concatenation recovers it. We therefore report concat as the construction and the bare residual as a diagnostic.

The few-class lineage tells the complementary half. On from-scratch CIFAR-10 the concatenation is a real gain on *both* currencies over its SupCon parent (probe +0.034, Euclidean +0.056, tied-Mahalanobis +0.015, top-1 tie), and the SupCon+SIGReg parent’s NPLM-residual child is the best-calibrated leakage-free CIFAR-10 space we measure (Euclidean AUC 0.758, per-event 0.12 against the parent’s 0.01). The residual adds probe where the parent already has geometry, and geometry where the parent already has the probe (Table 22).

The gain is not width. The concatenation is two full encoders, so the natural control is whether a single encoder of the same width would do as well. It does not: on cars/VISReg a 200-D SupCon head recovers +0.02 of the residual concat’s +0.15 probe gain and *loses* Euclidean AUC (0.675 vs. the parent’s 0.715) where the concat gains it (0.747); the concat beats the control by +0.12 probe, +0.08 Euclidean and +0.11 Mahalanobis, each five times the run-to-run variation. On Galaxy10 the probe gain is real on LeJEPa (+0.06 over the control) and a tie on DINO (+0.01), and on both backbones the wider single head is the better *geometry* space (Table 13). Width buys calibration cheaply; the residual buys the probe.

5.2 Composing with discovery: a 2×2

Because the residual is defined against the anchors, it can be built before or after the discovery loop. Writing A for the parent alone, B for the parent after discovery, C for the construction before

discovery, and D for both, the quantity of interest is not $D - A$ but the interaction

$$\Delta_{\text{int}} = (D - C) - (B - A), \quad (7)$$

which is zero for an additive stack and positive only when the two ingredients do more together than apart. We report it for per-event power at a nominal false-positive rate $\alpha = 0.05$, because that metric is bounded below by α : a parent at $A \approx \alpha$ has *no* ability to flag an individual novel sample, so a positive interaction there is a threshold crossing rather than an improvement.

The interaction is real, and it is narrow. On DTD with a DINO backbone, the residual construction gives $A = 0.040$, $B = 0.068$, $C = 0.022$, $D = 0.253$. Discovery alone is worth $+0.028$; the construction alone is worth -0.018 — it actively hurts — and composed they reach $6\times$ nominal, an interaction of $+0.203$. The concatenation behaves the same way ($+0.163$). Neither ingredient does anything alone. This is the strongest evidence we have that discovery and representation construction are not independent improvements.

Two apparent replications are not replications. The same protocol on Galaxy10 produces the two largest raw gains in the study, and the decomposition shows that neither is composition. With a DINO backbone the residual takes per-event power from 0.021 to 0.299, but $C = 0.281$ already: the construction does this *by itself*, and the interaction is $+0.003$. With a LeJEPa backbone the gain from 0.007 to 0.206 is almost entirely the discovery main effect ($+0.190$), with an interaction of $+0.008$ — and it occurs in a cell whose round-2 pool purity is exactly zero. That last case is worth stating plainly, because we had written it in advance as a falsifier: if the composition helps where purity is zero, then the mechanism we proposed for it — that discovery supplies a richer and *correct* anchor set, against which the residual becomes informative — cannot be what is happening there.

Building the residual *after* discovery makes the mechanism visible. Of the three cells with a before-discovery counterpart, the after-discovery concat raises the probe in all three and the geometry in two (DTD/DINO: probe $0.845 \rightarrow 0.898$, Euclidean $+0.10$, per-event $0.04 \rightarrow 0.23$; Galaxy10/VISReg: $+0.01$, $+0.13$, $+0.17$; on cars/VISReg the probe rises by $+0.05$ while the geometry is flat to slightly lower; Table 14). But in every one of the five cells, including the two whose round-1 pool purity was at or below 0.07 (one of which pseudo-labels only nine images), the images whose nearest anchor is a *discovered* one are 100% novel. The clustering step is noisy; the post-hoc nearest-anchor assignment is not. So the round-1 purity governs whether the *loop* compounds, not whether the anchors it planted sit on the novel class — which is why a construction defined against those anchors can help when the pool is poor, and why the LeJEPa cell above is not a counterexample to the anchors being right, only to the loop having needed them.

What we therefore claim. Across five cells, two constructions, and two variants, 12 of 20 interactions are positive, but most are small and the threshold crossings driven by interaction rather than by a main effect occur in a single cell with a single construction. We do not claim superadditivity as a general property. The claim the data supports is weaker and still useful: *discovery and the residual are substitutes more often than complements*. Whichever of the two a cell is missing, one of them supplies it — Galaxy10/LeJEPa needed discovery, Galaxy10/DINO needed the residual, DTD/DINO needed both — and a practitioner who runs only one of them will, on most cells, capture most of the available gain. Every number in this subsection is a single seed; the interaction is a difference of differences and inherits the variance of both its terms, so we treat it as a finding to be replicated rather than a headline.

5.3 Frozen-space discovery

Freeze the encoder, score the corpus by the density ratio of Section 4.2, and iterate only the anchors. Nothing in the representation moves; the loop’s only output is an enlarged anchor set. This is the construction the campaign’s single-holdout results converge on, for a reason that only the draw-resampled tables make visible.

The fine-tuning loop is backbone-dependent. On Galaxy10 — the one dataset none of our backbones has seen — the standard loop (encoder trains, distance pool) reaches a round-1 purity of only 0.02–0.04 on random single-holdout draws with a DINO backbone, 0.03–0.07 with LeJEPa

and 0.05–0.09 with VISReg, discovery from a 2% injected sample (Table 42; with the whole class present in the corpus the same numbers are 0.09–0.29). The archived alphabetical draw had reported purities of 0.46–0.50 on every backbone; across five random classes the same arm is 0.13–0.26. Whether “discovery works on Galaxy10” depends on which galaxy morphology is novel and which backbone embeds it, and no single-draw table can show that.

The frozen density-ratio pool is not. With the same parents frozen and the pool scored by the NP ratio, round-1 purity clears 0.15 on 143 of 144 draw×seed cells (95% CI [0.96, 1.00]), with cell means of 0.39–0.58 across every backbone and objective, a median of 0.58, and no round-2 collapse anywhere (Table 4); the single exception is at 0.10, and the draw-to-draw spread within a cell is large (sd 0.17). On the leakage-free CIFAR-10 lineage the same construction reaches 0.26–0.55 on all eight objectives, where the distance pool spans 0.00–0.46 with five objectives below 0.10 (Table 20). It costs nothing in the probe — the encoder does not move — and it removes the round-2 collapse that every fine-tuning run in Tables 42 and 45 shows: the collapse is the fine-tune destroying the pool, not the anchors. The construction does not even need the fine-tune: pooling the *pretrained* backbone’s features, with seen-class centroids as anchors and no training at all, reaches 0.56–0.60 purity on Galaxy10 and 0.26 on DTD on all 30 single-holdout draw cells — above the frozen fine-tuned head (0.39–0.58) and far above the fine-tuning loop (0.04–0.09 in the paper’s injected regime; 0.13–0.29 even with the whole held-out class present in the corpus), and above the fine-tuned default on DTD (0.21–0.22 injected). Fine-tuning on the seen classes improves the probe and the multi-holdout regime; it does not improve single-holdout discoverability, and on Galaxy10 it reduces it. Where the base rate is low the same construction is marginal rather than dead — flowers at 1/102 sits at 0.16–0.18, marginal on every draw — and where the novel class sits inside the seen manifold the pool is indistinguishable from chance regardless of rate (aircraft at 1/100: 0.01; cars at 1/196: 0.01; Table 15). That ordering — Galaxy10, DTD, flowers, aircraft, cars — is the regime boundary of Section 6 stated on one construction.

What “frozen” has to mean. A backbone with BatchNorm keeps adapting its running statistics in training mode even with every weight frozen, and on the CIFAR ResNet those statistics absorbed the discovery corpus, novel points included. Setting the backbone to evaluation mode changes the numbers: the round-2 rise of our earlier frozen archive (0.358 → 0.418) disappears (0.268 → 0.237), and a paired A/B on the same checkpoint attributes the round-2 rise entirely to the statistics adapting (0.237 → 0.299 with adaptation, ~30% embedding drift, no probe cost). That is a legitimate construction — corpus-adaptive normalisation uses no labels — and *corpus-adaptive normalisation* is its name in the code archive; but it is not “frozen”, and its transfer to LayerNorm backbones is a null result (−0.006 round-2 purity across 45 Galaxy10 draw cells). The frozen numbers in this paper are evaluation-mode numbers throughout.

6 Experiments

How to read the tables. The vocabulary of Section 3 — objective (arm), backbone, draw, seed, cell, and the single- versus multi-holdout and whole-class versus injected regimes — is used without further comment from here on. Distinguishing draw from seed matters more than it may appear, and Section 6 measures both: they turn out to differ by almost an order of magnitude, which determines how much of a single-run number can be believed.

The campaign is complete as reported; residual gaps are listed at the end of this section. Every table below states its own coverage, blanks are shown as “–” rather than dropped, and we mark which claims rest on a single seed. Datasets are ordered by *distance from pretraining* rather than alphabetically, because that ordering is what decides whether a result is evidence for discovery at all: Galaxy10 (Leung and Bovy, 2019; Walmsley et al., 2022) was never seen by any backbone we use, DTD (Cimpoi et al., 2014) is task-novel, cars/flowers/aircraft (Krause et al., 2013; Nilsback and Zisserman, 2008; Maji et al., 2013) are fine-grained but in-domain, and the CIFAR cells (Krizhevsky, 2009) exist in two lineages — a from-scratch one that we report, and a hub-pretrained one whose encoder already saw the held-out class and which we therefore treat as an ablation only.

objective	sup.	decodable			discoverable		
		probe	top-1	anchor	Eucl.	Maha.	per-event
SimCLR	n	0.799	0.292	0.242	0.443	0.385	0.01
VISReg/LeJEPa	n	0.797	0.239	0.190	0.499	0.382	0.03
NPLM (unsup.)	n	0.757	0.223	0.181	0.413	0.361	0.01
SupCon	y	0.940	0.608	0.545	0.233	0.279	0.00
SupCon+SIGReg (repulse)	y	0.823	0.360	0.346	0.340	0.320	0.00
NPLM-dist.+classwise SIGReg	y	0.898	0.507	0.463	0.525	0.411	0.03
SupCon+SIGReg ($\lambda=5$)	y	0.937	0.597	0.588	0.620	0.602	0.07
NPLM-dist. (sup.)	y	0.820	0.254	0.232	0.408	0.345	0.05

Table 1: Objectives on the leakage-free CIFAR-100 lineage (random init; the encoder never saw the held-out class). *Decodable* columns: linear-probe AUC for the held-out class (trained *with* its labels), top-1 accuracy on the seen classes, and nearest-anchor accuracy on the seen classes. *Discoverable* columns: novelty-ranking AUC by Euclidean and by tied-Mahalanobis distance, and per-event detection power at a 5% false-positive rate — the last is ≈ 0.05 for a space carrying no usable signal, so 0.00 means the space cannot flag an individual novel image at all. Adding SIGReg is a tie on the probe and on top-1 (a gap within run-to-run variation), and wins every discoverability column. *Coverage*: 8/8 objectives present; single holdout (class 4), one seed per cell, probe averaged over 3 probe restarts.

6.1 Objectives: the probe and the geometry come apart

Table 1 is the paper’s central measurement, on the lineage where it is safe to make it. Adding sliced-isotropy to a supervised contrastive objective is a tie on the linear probe and a tie on top-1 accuracy — and it more than doubles every discoverability column. A reader who ranked these eight objectives by probe accuracy would pick the one that scores 0.00 on per-event detection.

That is the dissociation stated at the start of the paper, measured on an encoder that was trained from random initialisation and never saw the class it is asked to find. It is also a *many-class* measurement. The same eight objectives trained from scratch on CIFAR-10 (Table 18) rank differently: there SupCon’s own geometry is the better one, the isotropy term improves neither currency, and the objective that carries both is the class-wise-marginal distance-NPLM. Three datasets now agree on the shape: SIGReg adds discoverability where classes are many and near, and is neutral-to-harmful where they are few and separable.

Seeing the unlabelled corpus is a dose, not a switch. Every objective above trains on the labelled classes only. The generalised category discovery (GCD) literature trains instead on the whole corpus, held-out images included, with an instance-level term on the unlabelled ones (Vaze et al., 2022a; Wen et al., 2023); scored on our battery under our protocol, that loss is the best-calibrated and highest-purity space we have on Galaxy10 with a SIGReg-pretrained backbone (tied Mahalanobis 0.85 vs. 0.71 for SupCon on LeJEPa, per-event power 0.40 vs. 0.05) — but only when the corpus contains the whole novel class. Table 2 varies that fraction: the gain is monotone in the dose, crosses SupCon between 20% and 50% presence on Galaxy10, and is absent at the $\leq 5\%$ contamination this paper is about. On DTD the plain GCD loss never leaves SupCon’s neighbourhood below full presence; the SIGReg marginal carries the effect there, and with four novel images in the corpus GCD+SIGReg is SupCon+SIGReg. At ten simultaneous novel classes (cars, aircraft, flowers) it ties or loses on nearly every currency — winning only tied Mahalanobis on flowers — and costs 0.04–0.28 of top-1. So the transductive term is the right construction when the whole novel class is available unlabelled, and changes nothing in the small-sample regime.

6.2 Stability: draws matter, seeds do not

Two variance questions decide how much of this campaign can be reported without qualification.

The first is the holdout *draw* — which class is held out. Table 3 reports five draws per cell. Pool purity is stable across draws for the best objective (sd 0.004–0.017) while the linear probe over the same draws swings by more than 0.1, so purity is the better-conditioned measurement of the two, not merely the more relevant one.

novel class	objective	probe	mahaT	per-event	purity
<i>Galaxy10 / LeJEPa</i>					
0%	SupCon	0.892±0.062	0.711±0.129	0.05±0.05	0.034±0.009
0%	SupCon+SIGReg ($\lambda=5$)	0.762±0.202	0.665±0.182	0.14±0.12	0.043±0.052
5%	GCD	0.800±0.150	0.647±0.033	0.17±0.25	0.029±0.008
5%	GCD+SIGReg	0.810±0.102	0.651±0.086	0.13±0.15	0.040±0.030
20%	GCD	0.857±0.100	0.695±0.076	0.19±0.28	0.050±0.057
20%	GCD+SIGReg	0.800±0.155	0.702±0.034	0.20±0.07	0.054±0.037
50%	GCD	0.882±0.110	0.771±0.099	0.31±0.30	0.105±0.094
50%	GCD+SIGReg	0.843±0.095	0.799±0.072	0.30±0.14	0.120±0.049
100%	GCD	0.899±0.098	0.853±0.064	0.40±0.29	0.131±0.086
100%	GCD+SIGReg	0.872±0.082	0.863±0.072	0.41±0.12	0.188±0.072
<i>DTD / DINO</i>					
0%	SupCon	0.883±0.044	0.628±0.086	0.16±0.15	0.028±0.014
0%	SupCon+SIGReg ($\lambda=5$)	0.787±0.041	0.732±0.060	0.42±0.11	0.211±0.017
5%	GCD	0.835±0.068	0.673±0.119	0.35±0.23	0.036±0.025
5%	GCD+SIGReg	0.837±0.114	0.747±0.070	0.44±0.19	0.119±0.018
20%	GCD	0.859±0.065	0.668±0.069	0.25±0.20	0.026±0.020
20%	GCD+SIGReg	0.901±0.059	0.797±0.093	0.38±0.23	0.159±0.046
50%	GCD	0.910±0.048	0.731±0.089	0.27±0.18	0.044±0.031
50%	GCD+SIGReg	0.917±0.038	0.868±0.062	0.53±0.19	0.206±0.039
100%	GCD	0.939±0.046	0.782±0.130	0.41±0.31	0.109±0.076
100%	GCD+SIGReg	0.926±0.048	0.900±0.097	0.56±0.23	0.234±0.039

Table 2: Seeing the unlabelled novel images during representation learning is a dose, not a switch. GCD’s representation loss (SimCLR over every image, held-out ones included with masked labels, plus SupCon on the labelled ones, 0.65:0.35; ‘+SIGReg’ adds our marginal at $\lambda=5$) scored on our battery as a function of the fraction of the held-out class present in the training corpus. The gain in calibration, per-event power and pool purity is monotone in the dose and absent at the contamination levels this paper is about ($\leq 5\%$); on DTD the plain GCD loss never leaves SupCon’s neighbourhood below 100% and it is the SIGReg marginal that carries the effect. *Coverage:* 20 (dose, objective) rows; single holdout, 5 draws each, one seed; ‘0%’ rows are the archived labelled-only arms on the same draws; the GCD arms train on the whole corpus with the held-out images unlabelled, a random fraction of them kept per dose; per-event power and purity are the POST grid at $f = 0.02$ (discovery from a 2% injected sample).

objective	DTD / DINO				DTD / LeJEPa			
	mean	sd	min	whole	mean	sd	min	whole
SimCLR	0.004	0.009	0.000	0.005	0.001	0.002	0.000	0.002
SimCLR+SIGReg	0.003	0.003	0.000	0.003	0.020	0.021	0.000	0.021
NPLM-bilinear	0.004	0.006	0.000	0.004	0.029	0.015	0.016	0.033
SupCon	0.028	0.014	0.011	0.028	0.068	0.080	0.011	0.073
SupCon+SIGReg ($\lambda=5$)	0.211	0.017	0.186	0.219	0.216	0.004	0.210	0.225
NPLM-dist. (sup.)	0.049	0.030	0.021	0.051	0.208	0.028	0.182	0.215

Table 3: Pool purity is stable across holdout draws, and only one objective is stable across *backbones*. Round-1 pool purity of discovery from a 2% injected sample of the held-out class: mean $\pm$ sd over draws, with the worst draw. The ‘whole’ column is the natural pass, in which the whole held-out class is present in the unlabelled bank — a different, easier regime, shown for reference only. SupCon+SIGReg holds its purity on both backbones; supervised distance-NPLM matches it on LeJEPa and loses a factor of four on DINO. *Coverage:* DTD, 5 holdout draws per base, single holdout; 12/12 (arm, base) pairs present. Purity read from the run logs (POST grid at $f = 0.02$; ‘whole’ = the natural pass).

The second is the *seed*. We specified this in advance as a falsification criterion: if seed-to-seed spread were comparable to draw-to-draw spread, every single-seed number in this paper would need a caveat. Table 4 settles it on 144 cells — $\text{sd}_{\text{seed}} = 0.021$ against $\text{sd}_{\text{draw}} = 0.174$, a factor of 8. The caveat that survives is about which class you hold out, not which seed you ran. We attach it to every single-seed result rather than dropping it.

Table 3 also carries a result we did not anticipate. Only one objective holds its purity across *both* backbones. SupCon+SIGReg reaches 0.211 on DINO and 0.216 on LeJEPa, and its worst

round	strata	mean Π	sd_{seed}	sd_{draw}	ratio
r1	9	0.516	0.021	0.174	$8.3\times$
r2	9	0.517	0.020	0.175	$8.9\times$

Table 4: The variance that matters is the holdout draw, not the seed. One-way decomposition of frozen-anchor round-1 purity, stratified within (cell, objective) so the seed term is measured at fixed draw. Seed-to-seed spread is an eighth of draw-to-draw spread in both rounds, so the single-seed purities elsewhere in this paper carry a caveat about which class was held out and not about which seed was run. *Coverage:* Galaxy10, 3 backbones $\times$ 3 objectives $\times$ 5 draws $\times$ 3 seeds (+1 archived) = 144 cells; variance stratified within (cell, objective). Round-level distribution summaries are not reported: the archive stores one summary for both rounds.

cell	child loss	used as	A	B	C	D	$B-A$	$C-A$	Δ_{int}
cars / VISReg	SupCon	bare	0.223	0.171	0.073	0.066	-0.052	-0.150	+0.045
cars / VISReg	SupCon	concat	0.223	0.171	0.228	0.174	-0.052	+0.005	-0.002
cars / VISReg	NPLM	bare	0.223	0.171	0.256	0.284	-0.052	+0.033	+0.080
cars / VISReg	NPLM	concat	0.223	0.171	0.263	0.282	-0.052	+0.040	+0.070
DTD / DINO	SupCon	bare	0.040	0.068	0.022	0.253	+0.028	-0.018	+0.203
DTD / DINO	SupCon	concat	0.040	0.068	0.045	0.235	+0.028	+0.005	+0.163
DTD / DINO	NPLM	bare	0.040	0.068	0.113	0.087	+0.028	+0.073	-0.053
DTD / DINO	NPLM	concat	0.040	0.068	0.077	0.130	+0.028	+0.037	+0.025
Galaxy10 / DINO	SupCon	bare	0.021	0.037	0.018	0.014	+0.015	-0.003	-0.020
Galaxy10 / DINO	SupCon	concat	0.021	0.037	0.021	0.018	+0.015	-0.001	-0.018
Galaxy10 / DINO	NPLM	bare	0.021	0.037	0.281	0.299	+0.015	+0.259	+0.003
Galaxy10 / DINO	NPLM	concat	0.021	0.037	0.091	0.142	+0.015	+0.069	+0.036
Galaxy10 / LeJEPA	SupCon	bare	0.007	0.196	0.001	0.002	+0.190	-0.005	-0.189
Galaxy10 / LeJEPA	SupCon	concat	0.007	0.196	0.001	0.012	+0.190	-0.005	-0.179
Galaxy10 / LeJEPA	NPLM	bare	0.007	0.196	0.009	0.021	+0.190	+0.002	-0.177
Galaxy10 / LeJEPA	NPLM	concat	0.007	0.196	0.008	0.206	+0.190	+0.001	+0.008
Galaxy10 / VISReg	SupCon	bare	0.311	0.419	0.243	0.457	+0.108	-0.068	+0.107
Galaxy10 / VISReg	SupCon	concat	0.311	0.419	0.295	0.466	+0.108	-0.015	+0.063
Galaxy10 / VISReg	NPLM	bare	0.311	0.419	0.333	0.471	+0.108	+0.022	+0.030
Galaxy10 / VISReg	NPLM	concat	0.311	0.419	0.353	0.446	+0.108	+0.043	-0.015

Table 5: Discovery $\times$ construction, per-event power at $\alpha=0.05$. Each row is one 2×2 : A is the parent space alone, B the parent after the discovery loop, C the construction built on the parent *without* discovery, and D both together. The two main effects are $B-A$ (discovery) and $C-A$ (construction). The residual child is trained under the loss in the second column and then either used alone (*bare*) or concatenated with its parent (*concat*). The interaction $\Delta_{int} = (D-C) - (B-A)$ is zero if the two simply add, and positive only if they do more together than apart. Bold marks $\Delta_{int} > 0.10$. Note that the two largest *totals* in the table (both Galaxy10) have near-zero interaction: they are main effects, not composition. *Coverage:* 5 cells $\times$ 2 constructions $\times$ 2 variants; 20 complete, 0 lacking a pre-discovery child. One seed per cell.

single draw (0.186) exceeds every other objective’s mean on DINO and all but supervised distance-NPLM’s on LeJEPA. Supervised distance-NPLM matches it on LeJEPA (0.208) and loses a factor of four on DINO (0.049) — the same objective, the same data, the same code, differing only in the pretrained backbone. Backbone-independence is a stronger and more useful property than winning on average, and it is not visible in any single-backbone table.

6.3 Constructions

Table 5 gives the 2×2 of Section 5 in full. The interaction is large in one cell, positive but small in most, and the two largest *totals* in the table are main effects rather than composition. We report all twenty rows rather than the favourable ones.

6.4 The regime boundary, on one construction

Table 6 runs the zero-training construction of Section 5.3 — pretrained features, seen-class centroids, density-ratio pool — on every transfer dataset at single holdout, five draws each. It orders the datasets more sharply than any fine-tuned table: Galaxy10 (0.56–0.60) and DTD (0.26) beat

dataset	rate	backbone	purity	min	fine-tuned loop, best objective
Galaxy10	1/10	DINO	0.598 ± 0.061	0.524	0.043 (nplm-sup-ft)
		LeJEPa	0.591 ± 0.040	0.543	0.068 (nplm-sup-ft)
		VISReg	0.556 ± 0.026	0.524	0.091 (supcon-ft)
DTD	1/47	DINO	0.257 ± 0.016	0.240	0.211 (ss-ft)
		LeJEPa	0.261 ± 0.014	0.246	0.216 (ss-ft)
		VISReg	0.267 ± 0.014	0.252	0.229 (ss-ft)
flowers	1/102	DINO	0.158 ± 0.006	0.151	—
		LeJEPa	0.177 ± 0.039	0.129	—
		VISReg	0.159 ± 0.016	0.144	—
aircraft	1/100	DINO	0.016 ± 0.010	0.003	—
		LeJEPa	0.010 ± 0.007	0.000	—
		VISReg	0.010 ± 0.008	0.000	—
cars	1/196	DINO	0.009 ± 0.008	0.000	—
		LeJEPa	0.007 ± 0.008	0.000	—
		VISReg	0.010 ± 0.006	0.002	—

Table 6: The regime boundary on one construction that needs no training. Round-1 pool purity of the pretrained trunk pooled by density ratio, single holdout, mean $\pm$ sd over five held-out classes, with the worst draw. Galaxy10 and DTD sit far above the best fine-tuned objective on every draw; flowers is marginal; aircraft and cars are dead although aircraft’s rate equals flowers’. *Coverage:* 75 cells; pretrained ViT-B/16 features, identity head, anchors = seen-class centroids, density-ratio pool, anchors-only iteration, round 1; five single-holdout draws per (dataset, backbone). Reference = the best objective of the exp-70 fine-tuning loop on the same draws, discovery from a 2% injected sample (paper regime, not the whole-class natural pass).

the best fine-tuned objective on the same draws on all 30 cells; flowers is marginal (0.16–0.18); aircraft and cars sit at the base rate (0.01). Flowers and aircraft have the same novel rate, so the rate is not what separates them: the pretrained backbone already places an unseen flower species away from the seen ones and an unseen aircraft variant among them. The base rate is necessary — cars at 1/196 sits at the base rate in both regimes — but the boundary is set jointly by the rate and by where the novel class sits in the pretrained space, which is what the earlier on-/off-manifold split measured with local intrinsic dimension. Two consequences for practice: on the datasets where discovery works, fine-tuning on the seen classes is not what makes it work (on Galaxy10 it lowers single-holdout purity from 0.60 to 0.04–0.09 in the paper’s injected regime, and to 0.13–0.29 even with the whole held-out class present); and whether a new dataset is discoverable can be read off the pretrained backbone before any training is done.

6.5 Where the label-free cut does not reach

Table 7 is a negative result about our own method. At a 1% novel rate on leakage-free CIFAR-100 feature banks, the label-free rule of Section 4.3 engages on 3 of 20 spaces, and where it engages the resulting purity is 0.046 to 0.101 — an enrichment of 5 to 10 over the 1% base rate, but far short of what the same cut admits. The spaces where it works share an NPLM critic (the class-wise-SIGReg arm and the NPLM residual children of a SupCon parent; the NPLM children of the SIGReg parent do not fire), so the rule should be read as conditional on the scorer, not as a general-purpose procedure. The ceiling algebra says a $20\times$ enrichment is available at this rate; the estimator, not the representation, is what fails to reach it.

7 When it does not work

The rate floor is a signal-efficiency limit, not an information limit. At a 1% novel rate on CIFAR-100 the pipeline as originally built — distance pool, 95th-percentile cut — reaches purity 0.03 on a ceiling of 0.20. The algebra of Section 4.3 says the ceiling is the cut’s, and the density-ratio scorer with a derived cut reaches 0.58 with the true labels and 0.39 label-free on the objectives that carry an NPLM critic on a SIGReg marginal. On the leakage-free lineage the label-free rule declines on 17 of 20 spaces at that rate (Table 7); the estimator, not the representation, is what stops

space	$\hat{b}/b$ (tv)	$\hat{b}/b$ (mass)	rule fires	q^*
nplm_cifar100	0.01	0.01	no	—
nplmcw_cifar100	1.04	0.49	yes	0.0277
nplmsd-res-nplm_(concat)_cifar100	0.15	0.08	no	—
nplmsd-res-nplm_(residual)_cifar100	0.07	0.04	no	—
nplmsd-res_(concat)_cifar100	0.17	0.14	no	—
nplmsd-res_(residual)_cifar100	0.37	0.28	no	—
nplmsd_cifar100	0.92	0.07	no	—
simclr_cifar100	0.09	0.09	no	—
ssig-res-nplm_(concat)_cifar100	0.18	0.04	no	—
ssig-res-nplm_(residual)_cifar100	0.43	0.36	no	—
ssig-res_(concat)_cifar100	0.04	0.03	no	—
ssig-res_(residual)_cifar100	0.01	0.01	no	—
ssig_cifar100	0.36	0.20	no	—
supcon-res-nplm_(concat)_cifar100	0.78	0.21	yes	0.0099
supcon-res-nplm_(residual)_cifar100	0.69	0.55	yes	0.0183
supcon-res_(concat)_cifar100	0.03	0.03	no	—
supcon-res_(residual)_cifar100	0.14	0.07	no	—
supcon_cifar100	0.00	0.01	no	—
supsig_cifar100	0.07	0.04	no	—
VISReg_cifar100	0.47	0.40	no	—

Table 7: The label-free cut is objective-specific, not universal. Ratio of estimated to true novel base rate at $b=0.01$; the rule engages only on spaces whose scorer is an NPLM critic (the class-wise SIGReg arm and the NPLM residual children of SupCon), and even then the resulting purity is only 0.046–0.101 (against a base rate of 0.01). Reported as a limit of the rule, not of the representations. *Coverage*: 20 spaces, clean CIFAR-100 banks, $b_{\text{true}}=0.01$; the rule engages on 3, and the best estimator is within $5\times$ on 8.

at the floor, and we have not fixed it. Three candidate fixes — swapping the estimator, a scale-invariant knee criterion, and injection calibration — all fail on real embeddings, because the bias is not a uniform scale (it grows as the cut widens). Lowering the clusterability constant is the one intervention that helps, and it is a tuning constant we report with its sweep rather than a correction. And the floor is not purely a rate: at the same 1%, flowers is discoverable from the pretrained backbone and aircraft is not (Table 6).

SIGReg does not help by making distance a log-likelihood. Forcing unit class-conditional width on demand is achievable and without downstream effect: the identity is satisfied, and nothing downstream improves. No space in the campaign reaches unit width class-conditionally (widths 0.13–0.79), and seen-class fidelity to the identity *anti*-correlates with novelty calibration. Five independent tests agree that repairing the identity is not the same as improving the space. The defensible claim is the one made in Section 4: SIGReg earns its place empirically, and it is what licenses the unit-variance clustering criterion.

The temperature basin is a landscape finding. At 100 classes a high softmax temperature with the marginal on lifts tied-Mahalanobis from 0.24 to ~ 0.51 and turns per-event detection on. No label-free or transductive criterion finds it — five legal selectors all pick the low temperature — it does not exist at 10 classes (the probe collapses to 0.65), and it costs 0.15–0.31 of supervised top-1. It is reported as a property of the loss landscape, not as a recipe.

The loop repairs weak spaces and damages strong ones. On the leakage-free lineages the fine-tuning loop is probe-positive on every objective whose pre-discovery probe is below 0.90 and probe-negative on those above 0.93, and it pulls every space toward one middle geometry: on CIFAR-100 it raises SupCon’s Euclidean AUC (0.23 $\rightarrow$ 0.30) and lowers SupCon+SIGReg’s (0.62 $\rightarrow$ 0.46) at pool purities of 0.00–0.01. A pool that empty cannot be what drives this; the fine-tune itself is a calibration repair, and it runs whether or not the pool contains anything. The one leakage-free exception locates the mechanism: on from-scratch CIFAR-10 with the class-wise marginal NPLM

— the only cell whose pool is substantially pure (0.43 with the whole class present, 0.12 from a 2% injection) — the loop raises an already-strong probe from 0.94 to 0.97 on every holdout draw. The rule is therefore not about the strength of the space but about what the pool supplies: an empty pool makes the fine-tune a calibration repair, and a pure one makes it discovery.

The NP loss transfers as a form, not as a target. Optimising the representation *through* a fitted NP kernel destroys the separation it measures, and warm-starting the kernel at class centroids is harmful: the centroids sit where the density ratio is one. The three roles of Section 4.2 share a functional form and a loss; they do not share learned content.

No-cost is a dataset-dependent claim. SupCon+SIGReg ties SupCon on supervised top-1 on both leakage-free CIFAR lineages (-0.011 and $+0.001$) and loses 0.035 to it on every one of ten Galaxy10 draws (Table 44). The residual concatenation ties its parent everywhere, including on every holdout draw of the leakage-free CIFAR lineages (top-1 within 0.006). The construction is no-cost; the objective is no-cost except on the one dataset the backbones had never seen.

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A Detailed results

This appendix reports the campaign in more detail than the main text, and reports it *as it stands* rather than waiting for completion. All tables in this paper are generated from the archived result files by a single script, and each carries a machine-written coverage line stating what is present and what is missing. Blanks are shown as “—”.

Throughout the appendices, “exp N ” refers to the numbered experiment index of the released code and log archive; the mapping from paper names to code names is Appendix E.

galaxy10, and how much of the difficulty is the procedure. Galaxy10 carries more weight than any other dataset here, because no backbone we use has seen it: DINO, LeJEPa and VISReg were all pretrained on natural images, and galaxy morphology is genuinely out of that distribution. It is also, for the same reason, the hardest cell in the study.

Table 8 reports it under both discovery procedures on the same holdout draws. Under natural discovery — a distance pool with the encoder fine-tuned and the whole held-out class present in the corpus (the whole-class pass) — round-1 pool purity sits at 0.09–0.29 across every objective and backbone, with sds that are frequently as large as the means, so the objectives are not separated by this measurement at all. Under the frozen-anchor density-ratio pool, with the same whole-class corpus, the same objectives on the same draws reach 0.39–0.58. In the paper’s own regime — discovery from a 2% injected sample of the held-out class, the left block of the table — the loop’s purity falls further, to 0.01–0.09; the frozen pool has not yet been measured at injected fractions, so that column has no counterpart.

That is a factor of three to five from changing the procedure and nothing else. It is the sharpest evidence in the paper for a claim we make throughout: failures that present as properties of the representation are frequently properties of the pipeline wrapped around it. On Galaxy10 specifically, the ordering of objectives under natural discovery is close to noise — the sds overlap almost completely — and the ordering under the frozen pool is both tighter and higher. We would not have found this by improving the encoder.

Note the coverage line: the frozen-pool arm of the study covers the three supervised objectives only, so nine of the eighteen cells in that half are absent rather than negative.

The per-dataset appendices that follow (B–D) put every measurement in one place. Each opens with a *summary grid*: one row per (objective, construction) with construction in {plain, +residual concat, +NPLM-residual concat}, and one column pair per metric for the space *before* and *after* the discovery loop — part (A) the probe and per-event geometry, part (B) the dataset-level SparKer, Mahalanobis and MMD power at $f=0.02$. A dash is a cell that was never measured, so the grids are also the coverage map of the campaign. The detail tables that follow each grid carry the remaining columns (accuracy, LID, pools, draws). This section keeps the cross-cutting tables.

Constructions on the leakage-free backbones. Table 9 gives every construction on the from-scratch CIFAR-100 parents. The pattern behind the summary in Section 5 is visible directly: the bare residual usually costs probe accuracy, the concatenation recovers it, and the detection gain survives in both. The CIFAR-10 counterpart is Table 22; the width control and the after-discovery construction are Tables 13 and 14.

Multi-seed cells. Most cells in this campaign are single-seed, for cost reasons. Table 10 reports one of the exceptions, three seeds on the best-performing CIFAR-10 configurations. Seed spread there is small — ± 0.001 on the best post-discovery probe and ± 0.002 on its Euclidean AUC — which is consistent with the much larger variance decomposition of Table 4. Note also that the discovery loop *reduces* the probe for `supcon+np1m` ($0.954 \rightarrow 0.930$) in all three seeds, which is the seeded version of a pattern we observe throughout: the loop repairs weak spaces and damages strong ones.

Known gaps. The campaign is complete; what remains open is the following. The 2×2 of Table 5 is single-seed throughout, and its one large interaction rests on a single cell; the multi-holdout regime is run only on the label-rich datasets; the leakage-free CIFAR shortlist and its residual constructions carry four holdout draws each (Tables 21 and 29) while the remaining four objectives of the cube are one holdout each; the frozen density-ratio pool has not been measured at injected fractions; flowers, cars and aircraft have no single-holdout fine-tuned parents, so on those datasets the training-free construction of Tables 6 and 15 is the only single-holdout evidence (it is also the strongest); and the frozen-space construction of Section 5.3 is quoted from evaluation-mode runs after a BatchNorm fix invalidated the earlier archive, so its older numbers do not reproduce and are not used.

objective	loop, 2% injected			loop, whole class present			frozen-anchor pool		
	DINO	LeJEPA	VISReg	DINO	LeJEPA	VISReg	DINO	LeJEPA	VISReg
SimCLR	0.012 $\pm$ 0.015	0.012 $\pm$ 0.018	0.005 $\pm$ 0.007	0.104 $\pm$ 0.100	0.112 $\pm$ 0.112	0.093 $\pm$ 0.070	–	–	–
SimCLR+SIGReg	0.014 $\pm$ 0.016	0.037 $\pm$ 0.052	0.012 $\pm$ 0.009	0.100 $\pm$ 0.099	0.123 $\pm$ 0.188	0.110 $\pm$ 0.086	–	–	–
NPLM-bilinear	0.010 $\pm$ 0.010	0.038 $\pm$ 0.022	0.022 $\pm$ 0.023	0.104 $\pm$ 0.089	0.102 $\pm$ 0.057	0.117 $\pm$ 0.103	–	–	–
SupCon	0.017 $\pm$ 0.023	0.034 $\pm$ 0.009	0.091 $\pm$ 0.049	0.103 $\pm$ 0.154	0.132 $\pm$ 0.095	0.290 $\pm$ 0.075	0.465 $\pm$ 0.158	0.575 $\pm$ 0.185	0.574 $\pm$ 0.180
SupCon+SIGReg ($\lambda=5$)	0.017 $\pm$ 0.014	0.043 $\pm$ 0.052	0.053 $\pm$ 0.058	0.092 $\pm$ 0.083	0.182 $\pm$ 0.223	0.146 $\pm$ 0.047	0.459 $\pm$ 0.143	0.516 $\pm$ 0.186	0.554 $\pm$ 0.169
NPLM-dist. (sup.)	0.043 $\pm$ 0.052	0.068 $\pm$ 0.066	0.089 $\pm$ 0.070	0.131 $\pm$ 0.043	0.190 $\pm$ 0.153	0.262 $\pm$ 0.119	0.393 $\pm$ 0.147	0.569 $\pm$ 0.179	0.543 $\pm$ 0.185

Table 8: Galaxy10, the dataset no backbone has seen, under two discovery procedures. Round-1 pool purity, mean $\pm$ sd over draws. Left: the fine-tuning loop’s distance pool, discovery from a 2% injected sample of the held-out class (the paper’s regime). Middle: the same loop’s natural pass with the whole held-out class present in the bank — a different, easier regime, for reference. Right: the frozen-anchor density-ratio pool, also with the whole class present in the bank (its injected-sample counterpart is not yet measured), so the like-for-like comparison is middle vs right: same dataset, same draws, a factor of three to five in purity from the procedure alone. The left block shows how much harder the paper’s own regime is for the loop. *Coverage*: Galaxy10, 3 backbones; fine-tuning loop over 5 draws (18/18 cells present, injected and whole-class passes from the same runs), frozen-anchor pool over 5 draws $\times$ 3 seeds (9/18 present — that arm of the study covers the three supervised objectives only).

construction	probe	Δ	Eucl. AUC	Δ	per-event
<i>parent: SupCon</i>					
(parent)	0.940	–	0.233	–	0.00
residual child (SupCon), bare	0.869	-0.071	0.607	+0.374	0.05
residual child (SupCon), concat	0.942	+0.002	0.521	+0.288	0.03
residual child (NPLM), bare	0.882	-0.058	0.291	+0.057	0.00
residual child (NPLM), concat	0.947	+0.007	0.227	-0.006	0.00
<i>parent: SupCon+SIGReg ($\lambda=5$)</i>					
(parent)	0.937	–	0.620	–	0.07
residual child (SupCon), bare	0.860	-0.077	0.604	-0.017	0.08
residual child (SupCon), concat	0.949	+0.012	0.648	+0.028	0.08
residual child (NPLM), bare	0.900	-0.037	0.464	-0.156	0.01
residual child (NPLM), concat	0.904	-0.033	0.575	-0.046	0.04
<i>parent: NPLM-dist. (sup.)</i>					
(parent)	0.820	–	0.408	–	0.05
residual child (SupCon), bare	0.834	+0.014	0.524	+0.116	0.01
residual child (SupCon), concat	0.867	+0.047	0.521	+0.114	0.01
residual child (NPLM), bare	0.827	+0.006	0.527	+0.119	0.06
residual child (NPLM), concat	0.849	+0.029	0.516	+0.108	0.04

Table 9: All constructions on the leakage-free CIFAR-100 trunks. Δ columns are against that parent. The bare residual generally loses probe accuracy and the concatenation recovers it; the detection gain is large and nearly free. *Coverage:* 3 parents $\times$ 4 constructions on the from-scratch CIFAR-100 trunks; one seed per cell.

space	probe (pre)	probe (post)	Eucl. AUC	Maha.	seeds
nplmsup+nplm	0.888 $\pm$ 0.010	0.917 $\pm$ 0.004	0.686 $\pm$ 0.016	0.752 $\pm$ 0.025	3
nplmsup $\rightarrow$ res	0.871 $\pm$ 0.015	0.879 $\pm$ 0.017	0.448 $\pm$ 0.015	0.621 $\pm$ 0.045	3
sup+nplm	0.926 $\pm$ 0.012	0.979 $\pm$ 0.001	0.794 $\pm$ 0.009	0.729 $\pm$ 0.044	3
sup $\rightarrow$ res-nplm	0.930 $\pm$ 0.012	0.983 $\pm$ 0.001	0.834 $\pm$ 0.002	0.759 $\pm$ 0.012	3
supcon+nplm	0.954 $\pm$ 0.002	0.930 $\pm$ 0.009	0.766 $\pm$ 0.005	0.723 $\pm$ 0.016	3

Table 10: Multi-seed CIFAR-10 champions. Mean $\pm$ sd over seeds. Seed spread here is small, consistent with Table 4. *Coverage:* CIFAR-10, 5 spaces, up to 3 seeds; one of the few multi-seed cells in the campaign.

space	probe pre	probe post	eucl	mahaT	SpK post	MMD post	n
<i>100-D (50+50), classwise-$\lambda 5$, 3 seeds</i>							
nplmsup+nplm	0.931 $\pm$ 0.008	0.938 $\pm$ 0.004	0.502 $\pm$ 0.112	0.434 $\pm$ 0.093	0.300 $\pm$ 0.075	0.807 $\pm$ 0.094	3
nplmsup $\rightarrow$ res	0.913 $\pm$ 0.008	0.926 $\pm$ 0.007	0.437 $\pm$ 0.004	0.341 $\pm$ 0.024	0.147 $\pm$ 0.038	0.300 $\pm$ 0.145	3
sup+nplm	0.928 $\pm$ 0.007	0.926 $\pm$ 0.007	0.525 $\pm$ 0.016	0.416 $\pm$ 0.076	0.227 $\pm$ 0.159	0.340 $\pm$ 0.102	3
sup $\rightarrow$ res-nplm	0.929 $\pm$ 0.003	0.925 $\pm$ 0.010	0.516 $\pm$ 0.018	0.385 $\pm$ 0.024	0.313 $\pm$ 0.232	0.440 $\pm$ 0.142	3
supcon+nplm	0.954 $\pm$ 0.008	0.951 $\pm$ 0.005	0.375 $\pm$ 0.044	0.403 $\pm$ 0.079	0.333 $\pm$ 0.050	0.800 $\pm$ 0.059	3
<i>32-D (16+16), 3 seeds</i>							
nplmsup+nplm	0.901 $\pm$ 0.009	0.902 $\pm$ 0.002	0.563 $\pm$ 0.026	0.505 $\pm$ 0.011	0.207 $\pm$ 0.047	0.667 $\pm$ 0.034	3
nplmsup $\rightarrow$ res	0.839 $\pm$ 0.006	0.866 $\pm$ 0.009	0.480 $\pm$ 0.026	0.402 $\pm$ 0.011	0.173 $\pm$ 0.034	0.500 $\pm$ 0.188	3
sup+nplm	0.897 $\pm$ 0.009	0.867 $\pm$ 0.006	0.576 $\pm$ 0.007	0.414 $\pm$ 0.012	0.087 $\pm$ 0.025	0.447 $\pm$ 0.082	3
sup $\rightarrow$ res-nplm	0.819 $\pm$ 0.024	0.835 $\pm$ 0.028	0.516 $\pm$ 0.019	0.347 $\pm$ 0.016	0.173 $\pm$ 0.025	0.500 $\pm$ 0.016	3
supcon+nplm	0.941 $\pm$ 0.001	0.935 $\pm$ 0.004	0.456 $\pm$ 0.022	0.445 $\pm$ 0.025	0.333 $\pm$ 0.168	0.753 $\pm$ 0.105	3

Table 11: Multi-seed CIFAR-100 champions (hub lineage). Power columns at $f=0.02$, post-discovery. *Coverage:* CIFAR-100, two exp-59 configurations, 6 result files; hub-pretrained trunk (ablation lineage).

space	probe	acc	eucl	mahaT
<i>CIFAR-10, 32d, 5 paired seeds</i>				
parent	0.939±0.002	0.869±0.007	0.722±0.019	0.667±0.026
res_concat	0.955±0.002	0.856±0.004	0.567±0.020	0.541±0.026
res-nplm_concat	0.945±0.002	0.873±0.007	0.717±0.023	0.593±0.026
<i>CIFAR-100, 100d, 5 paired seeds</i>				
parent	0.947±0.010	0.574±0.005	0.385±0.033	0.374±0.021
res_concat	0.952±0.008	0.534±0.004	0.607±0.020	0.494±0.013
res-nplm_concat	0.955±0.006	0.576±0.003	0.339±0.035	0.307±0.014

Table 12: Hub-lineage residual concats, 5 paired seeds (exp 75; ablation lineage). Coverage: exp 75, both CIFAR datasets, 5 seeds; hub-pretrained trunk (saw the held-out class).

cell	metric	parent (100)	control (200)	concat (100+100)	concat−control	verdict
cars-VISReg	probe	0.707	0.727±0.014	0.855	+0.116	concat
	eucl	0.715	0.675±0.009	0.747	+0.080	concat
	mahaT	0.661	0.591±0.018	0.678	+0.107	concat
Galaxy10-DINO	probe	0.938	0.944	0.955	+0.011	TIE
	eucl	0.666	0.652	0.559	-0.092	control
	mahaT	0.786	0.777	0.762	-0.014	TIE
Galaxy10-LeJEPA	probe	0.910±0.007	0.917	0.971±0.004	+0.058	concat
	eucl	0.690±0.036	0.734	0.372±0.063	-0.446	control
	mahaT	0.741±0.022	0.799	0.661±0.026	-0.174	control

Table 13: Width-matched control for the residual concat (exp 134a). A same-width non-residual head recovers little of the flagship probe gain and loses geometry on cars/VISReg; on Galaxy10 the probe gain is a tie on DINO. Coverage: 3 cells; control = the parent objective with a 200-D head (no residual); concat = 100+100 residual concat; TIE unless the gap clears $\max(0.017, sd_a + sd_b)$.

Table 14: Residual built after discovery vs. before (exp 134c). The pseudo-labelled child corpus is 100% novel in every cell, including where the pool purity is very low. *Coverage:* 5 cells, archived holdout draw, seed 0; parent = SupCon fine-tune; archived = exp-71 pre-discovery order.

concat	probe (pre)	probe (post)	eucl (pre)	eucl (post)	mahaT (pre)	mahaT (post)	per-ev (pre)	per-ev (post)
<i>cars_visreg</i> : round-1 purity 0.206, 217 pseudo-labelled at purity 1.00								
supcon-ft→res	0.801	0.847	0.705	0.692	0.613	0.607	0.23	0.17
supcon-ft→res-nplm	0.855	0.872	0.747	0.744	0.678	0.675	0.26	0.28
<i>DTD_dino</i> : round-1 purity 0.129, 511 pseudo-labelled at purity 1.00								
supcon-ft→res	0.845	0.898	0.683	0.781	0.604	0.780	0.04	0.23
supcon-ft→res-nplm	0.832	0.864	0.733	0.764	0.648	0.690	0.08	0.13
<i>Galaxy10_dino</i> : round-1 purity 0.070, 64 pseudo-labelled at purity 1.00								
supcon-ft→res	–	0.949	–	0.595	–	0.746	–	0.02
supcon-ft→res-nplm	–	0.938	–	0.728	–	0.782	–	0.14
<i>Galaxy10_lejepa</i> : round-1 purity 0.059, 9 pseudo-labelled at purity 1.00								
supcon-ft→res	–	0.971	–	0.432	–	0.740	–	0.01
supcon-ft→res-nplm	–	0.943	–	0.739	–	0.797	–	0.21
<i>Galaxy10_visreg</i> : round-1 purity 0.481, 108 pseudo-labelled at purity 1.00								
supcon-ft→res	0.964	0.974	0.676	0.807	0.868	0.918	0.30	0.47
supcon-ft→res-nplm	0.946	0.968	0.744	0.820	0.839	0.895	0.35	0.45

Table 15: Frozen-backbone pooling with no training at all, on the pretrained backbone features: distance vs density-ratio pool, multi- vs single-holdout regime; round-1 and round-2 purity with the worst draw. *Coverage*: 90 cell files (multi-holdout default draw, and single-holdout draws 0–4); pretrained ViT-B/16 features, identity head, anchors = seen centroids, no fine-tune.

cell	multi-holdout						single holdout (draws)					
	dist			np			dist			np		
	r1	r2	min	r1	r2	min	r1	r2	min	r1	r2	min
Galaxy10/DINO	0.036	0.158	0.036	0.615	0.630	0.615	0.164±0.022	0.128±0.002	0.146	0.598±0.061	0.593±0.061	0.524
Galaxy10/LeJEPa	0.024	0.101	0.024	0.585	0.600	0.585	0.182±0.012	0.113±0.022	0.167	0.591±0.040	0.586±0.044	0.543
Galaxy10/VISReg	0.036	0.059	0.036	0.581	0.577	0.581	0.230±0.012	0.153±0.026	0.216	0.556±0.026	0.574±0.041	0.524
DTD/DINO	0.668	0.383	0.668	0.611	0.624	0.611	0.105±0.020	0.074±0.025	0.080	0.257±0.016	0.256±0.027	0.240
DTD/LeJEPa	0.529	0.351	0.529	0.507	0.507	0.507	0.041±0.024	0.037±0.017	0.021	0.261±0.014	0.246±0.038	0.246
DTD/VISReg	0.580	0.295	0.580	0.526	0.552	0.526	0.015±0.014	0.013±0.010	0.000	0.267±0.014	0.245±0.027	0.252
flowers/DINO	0.562	0.319	0.562	0.495	0.446	0.495	0.098±0.041	0.051±0.033	0.038	0.158±0.006	0.154±0.010	0.151
flowers/LeJEPa	0.338	0.287	0.338	0.506	0.521	0.506	0.052±0.025	0.038±0.015	0.029	0.177±0.039	0.150±0.014	0.129
flowers/VISReg	0.303	0.193	0.303	0.483	0.575	0.483	0.046±0.038	0.034±0.020	0.010	0.159±0.016	0.160±0.020	0.144
cars/DINO	0.070	0.068	0.070	0.104	0.085	0.104	0.003±0.004	0.003±0.002	0.000	0.009±0.008	0.005±0.003	0.000
cars/LeJEPa	0.054	0.045	0.054	0.074	0.104	0.074	0.004±0.002	0.003±0.002	0.000	0.007±0.008	0.011±0.007	0.000
cars/VISReg	0.068	0.061	0.068	0.079	0.119	0.079	0.004±0.004	0.003±0.002	0.000	0.010±0.006	0.005±0.003	0.002
aircraft/DINO	0.176	0.130	0.176	0.157	0.437	0.157	0.029±0.018	0.028±0.018	0.003	0.016±0.010	0.021±0.022	0.003
aircraft/LeJEPa	0.123	0.083	0.123	0.088	0.180	0.088	0.014±0.013	0.017±0.013	0.000	0.010±0.007	0.009±0.008	0.000
aircraft/VISReg	0.138	0.085	0.138	0.125	0.208	0.125	0.015±0.012	0.012±0.009	0.000	0.010±0.008	0.009±0.008	0.000

B CIFAR-10 and CIFAR-100, leakage-free lineage: every measurement

All eight objectives of the design cube, trained from random initialisation on the seen classes of each dataset (ResNet-20, 100-D head, 200 epochs, holdout class 4), with the full battery on the frozen space, the frozen pools, the label-free cut, and the fine-tuning loop. CIFAR-10 holds out one class in ten ($b = 0.10$); CIFAR-100 one in a hundred ($b = 0.01$). Every number is seed 0; the shortlist objectives and the residual constructions additionally carry four holdout draws each (Tables 21, 29, 23 and 31). Reading the two datasets side by side is the point: the objective ranking, the discoverability of the frozen space, and the response to the loop all flip between them. In the grids, “post” for a plain space is the fine-tuning loop of exp 68 (encoder trains, distance pool, two rounds); discovery on the scratch residual concats has not been run.

B.1 CIFAR-10, leakage-free

Table 16: CIFAR-10, leakage-free lineage: every (objective, construction) $\times$ (pre, post). (A) Probe and per-event geometry. *Coverage:* 8/8 objectives, from random init, 100-D, holdout 4 (single holdout, $b = 0.10$), seed 0. ‘post’ = the exp-68 fine-tuning loop (distance pool, 2 rounds), discovery from a 2% injected sample; per-fraction post geometry present for 8/8 objectives (‘–’ otherwise; the whole-class natural pass is in the pools table). Residual concats: parents SupCon / SupCon+SIGReg / NPLM-dist. only (exp 137); discovery on scratch concats not run.

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
<i>SupCon</i>								
plain	0.905	0.898	0.514	0.706	0.552	0.537	0.000	0.004
+res concat	0.939	–	0.571	–	0.567	–	0.001	–
+res-nplm concat	0.916	–	0.576	–	0.585	–	0.000	–
<i>SupCon+SIGReg ($\lambda 5$)</i>								
plain	0.884	0.921	0.439	0.689	0.454	0.363	0.013	0.270
+res concat	0.932	–	0.407	–	0.383	–	0.016	–
+res-nplm concat	0.919	–	0.571	–	0.544	–	0.054	–
<i>NPLM-dist. +cw SIGReg</i>								
plain	0.921	0.928	0.729	0.661	0.559	0.488	0.194	0.381
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>SupCon+SIGReg (repulse)</i>								
plain	0.871	0.861	0.667	0.660	0.582	0.518	0.121	0.194
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>								
plain	0.719	0.900	0.692	0.556	0.659	0.501	0.123	0.097
+res concat	0.876	–	0.378	–	0.459	–	0.017	–
+res-nplm concat	0.842	–	0.610	–	0.543	–	0.030	–
<i>SimCLR</i>								
plain	0.808	0.907	0.227	0.493	0.224	0.410	0.003	0.022
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>VISReg/LeJEPA</i>								
plain	0.832	0.882	0.395	0.472	0.377	0.363	0.010	0.017
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>NPLM (unsup.)</i>								
plain	0.799	0.845	0.472	0.498	0.468	0.462	0.052	0.068
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–

Table 17: CIFAR-10, leakage-free lineage: every (objective, construction) $\times$ (pre, post). (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* 8/8 objectives, from random init, 100-D, holdout 4 (single holdout, $b = 0.10$), seed 0. ‘post’ = the exp-68 fine-tuning loop (distance pool, 2 rounds), discovery from a 2% injected sample; per-fraction post geometry present for 8/8 objectives (‘–’ otherwise; the whole-class natural pass is in the pools table). Residual concats: parents SupCon / SupCon+SIGReg / NPLM-dist. only (exp 137); discovery on scratch concats not run.

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
<i>SupCon</i>						
plain	0.88	0.22	0.14	0.34	0.84	0.16
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>SupCon+SIGReg ($\lambda 5$)</i>						
plain	0.16	0.12	0.00	0.04	0.46	0.12
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>NPLM-dist. +cw SIGReg</i>						
plain	0.38	0.22	0.20	0.52	0.48	0.34
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>SupCon+SIGReg (repulse)</i>						
plain	0.36	0.42	0.56	0.32	0.28	0.20
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>						
plain	0.16	0.16	0.22	0.38	0.16	0.00
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>SimCLR</i>						
plain	0.22	0.08	0.00	0.04	0.44	0.08
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>VISReg/LeJEPa</i>						
plain	0.04	0.00	0.12	0.10	0.08	0.02
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>NPLM (unsup.)</i>						
plain	0.04	0.02	0.02	0.20	0.02	0.00
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–

objective	probe	top-1	acc	supAUC	eucl	mahaT	mahaPC	lid	per-ev	probe sd
SupCon	0.905	0.896	0.741	0.915	0.514	0.552	0.646	0.703	0.00	0.002
SupCon+SIGReg ($\lambda=5$)	0.884	0.897	0.894	0.989	0.439	0.454	0.390	0.518	0.01	0.001
NPLM-dist.+classwise SIGReg	0.921	0.898	0.896	0.989	0.729	0.559	0.589	0.248	0.19	0.000
SupCon+SIGReg (repulse)	0.871	0.802	0.803	0.963	0.667	0.582	0.649	0.616	0.12	0.000
NPLM-dist. (sup.)	0.719	0.827	0.799	0.969	0.692	0.659	0.646	0.376	0.12	0.022
SimCLR	0.808	0.691	0.638	0.919	0.227	0.224	0.353	0.574	0.00	0.002
VISReg/LeJEPA	0.832	0.697	0.638	0.922	0.395	0.377	0.515	0.621	0.01	0.003
NPLM (unsup.)	0.799	0.485	0.435	0.824	0.472	0.468	0.473	0.484	0.05	0.001

Table 18: CIFAR-10 leakage-free lineage: the frozen-space battery. probe = holdout-vs-rest linear AUC; top-1 = supervised linear probe on seen classes; acc = nearest-centroid; supAUC = macro OvR AUC of the proto posterior; eucl / mahaT / mahaPC = novelty AUC of min-centroid distance, tied and per-class Mahalanobis; lid = local-intrinsic-dimension AUC; per-ev = per-event power at $\alpha=0.05$. *Coverage:* 8/8 objectives; CIFAR-10 from random init, 100-D, holdout class 4 (single holdout, $b = 0.10$), seed 0; probe and top-1 over 3 probe restarts.

objective	SparKer			Mahalanobis			MMD		
	0.01	0.02	0.1	0.01	0.02	0.1	0.01	0.02	0.1
SupCon	0.48	0.88	1.00	0.16	0.14	0.84	0.32	0.84	1.00
SupCon+SIGReg ($\lambda=5$)	0.06	0.16	1.00	0.00	0.00	0.00	0.18	0.46	1.00
NPLM-dist.+classwise SIGReg	0.12	0.38	1.00	0.10	0.20	0.94	0.18	0.48	1.00
SupCon+SIGReg (repulse)	0.08	0.36	1.00	0.24	0.56	1.00	0.08	0.28	1.00
NPLM-dist. (sup.)	0.14	0.16	1.00	0.10	0.22	0.92	0.08	0.16	1.00
SimCLR	0.10	0.22	1.00	0.02	0.00	0.00	0.22	0.44	1.00
VISReg/LeJEPa	0.04	0.04	1.00	0.06	0.12	0.36	0.04	0.08	1.00
NPLM (unsup.)	0.06	0.04	0.88	0.00	0.02	0.02	0.04	0.02	0.92

Table 19: CIFAR-10 leakage-free lineage: dataset-level detection power on the frozen space, before any discovery. Signal fraction f of held-out images injected into a seen-class test sample. *Coverage*: 8/8 objectives; annealed- σ SparKer, parametric Mahalanobis and MMD two-sample power at $\alpha=0.05$, $N_D=5000$, 200 null / 50 signal toys; fractions 0.01, 0.02, 0.1.

objective	frozen pools				legal cut		loop, 2% injected			loop, whole class		
	dist r1	np r1	np r2	np+BN r2	$n_{\min}=10$	$n_{\min}=30$	purity	probe	mahaT	purity	probe	pre→post
SupCon	0.000	0.547	0.550	0.571	0.731	0.722	0.000	0.898	0.537	0.000	0.907→0.912	
SupCon+SIGReg ($\lambda=5$)	0.029	0.396	0.385	0.441	0.812	0.738	0.005	0.921	0.363	0.029	0.886→0.896	
NPLM-dist.+classwise SIGReg	0.461	0.485	0.539	0.480	0.757	0.777	0.140	0.928	0.488	0.461	0.920→0.954	
SupCon+SIGReg (repulse)	0.450	0.354	0.297	0.370	0.380	ref.	0.132	0.861	0.518	0.450	0.870→0.875	
NPLM-dist. (sup.)	0.285	0.392	0.271	0.448	0.429	0.442	0.072	0.900	0.501	0.285	0.778→0.908	
SimCLR	0.011	0.443	0.475	0.434	0.848	0.871	0.002	0.907	0.410	0.011	0.814→0.905	
VISReg/LeJEPa	0.017	0.345	0.290	0.253	0.645	0.583	0.002	0.882	0.363	0.017	0.831→0.870	
NPLM (unsup.)	0.091	0.264	0.323	0.285	0.376	0.408	0.017	0.845	0.462	0.091	0.801→0.838	

Table 20: CIFAR-10 leakage-free lineage: pools and discovery. Round-1 purity of the frozen distance and density-ratio (np) pools, the np pool’s round 2 with BN frozen and adapting, the label-free cut, then the fine-tuning loop for *discovery from a 2% injected sample* of the held-out class (the paper’s regime): round-1 purity, post probe and post tied Mahalanobis. The last two columns are the loop’s natural pass, in which the whole held-out class is present in the unlabelled bank — a different, easier regime, shown for reference only. *Coverage:* 8/8 objectives; frozen pools = trunk frozen (BN in eval unless ‘BN adapt’), 2×2 -epoch anchor rounds; legal cut at $n_{\min} = 10$ and 30 (‘ref.’ = the rule declined); loop = exp-68 fine-tuning loop, distance pool, 2×5 epochs. Injected-sample probe/mahaT present for 8/8 objectives (‘-’ = npz predates the per-fraction keys).

objective	draws	pre-discovery				frozen	loop		
		probe	top-1	mahaT	per-ev	purity	purity 2%	purity whole	probe post whole
SupCon	4	0.930±0.017	0.892±0.003	0.587±0.078	0.011±0.020	0.588±0.031	0.003±0.005	0.018±0.031	0.924±0.016
SupCon+SIGReg ($\lambda=5$)	4	0.912±0.020	0.891±0.004	0.416±0.040	0.019±0.014	0.392±0.085	0.007±0.005	0.042±0.030	0.925±0.017
NPLM-dist. (sup.)	4	0.788±0.079	0.815±0.013	0.685±0.047	0.165±0.057	0.408±0.046	0.082±0.018	0.320±0.059	0.937±0.032
NPLM-dist.+classwise SIGReg	4	0.938±0.018	0.891±0.007	0.624±0.077	0.207±0.033	0.539±0.032	0.123±0.010	0.430±0.020	0.972±0.013

Table 21: CIFAR-10 from scratch, across holdout draws. Pre-discovery probe, top-1, tied Mahalanobis and per-event power; frozen density-ratio pool purity; the fine-tuning loop’s round-1 purity for discovery from a 2% injected sample and, for reference, with the whole held-out class present in the bank, with the whole-class post probe. *Coverage:* CIFAR-10 from scratch, single holdout, holdouts [4, 7, 8, 9] (the archived class plus random draws), one seed per draw; objectives with fewer than two draws omitted; injected-sample purity from the POST-grid logs (‘-’ = run predates it).

space	probe	top-1	acc	eucl	mahaT	lid	per-ev
<i>parent: SupCon</i>							
(parent)	0.905	0.896	0.741	0.514	0.552	0.703	0.00
res (residual)	0.902	0.879	0.864	0.650	0.602	0.622	0.02
res (concat)	0.939	0.899	0.815	0.571	0.567	0.678	0.00
res-nplm (residual)	0.892	0.889	0.872	0.659	0.664	0.421	0.05
res-nplm (concat)	0.916	0.902	0.847	0.576	0.585	0.635	0.00
<i>parent: SupCon+SIGReg ($\lambda=5$)</i>							
(parent)	0.884	0.897	0.894	0.439	0.454	0.518	0.01
res (residual)	0.878	0.662	0.583	0.380	0.377	0.551	0.01
res (concat)	0.932	0.896	0.889	0.407	0.383	0.513	0.02
res-nplm (residual)	0.888	0.885	0.872	0.758	0.617	0.566	0.12
res-nplm (concat)	0.919	0.897	0.896	0.571	0.544	0.529	0.05
<i>parent: NPLM-dist. (sup.)</i>							
(parent)	0.719	0.827	0.799	0.692	0.659	0.376	0.12
res (residual)	0.851	0.683	0.606	0.371	0.356	0.552	0.02
res (concat)	0.876	0.835	0.648	0.378	0.459	0.595	0.02
res-nplm (residual)	0.817	0.754	0.560	0.480	0.426	0.653	0.02
res-nplm (concat)	0.842	0.832	0.799	0.610	0.543	0.576	0.03

Table 22: CIFAR-10 leakage-free lineage: every residual construction, with the supervised top-1 that the main text’s no-cost claim rests on. *Coverage:* 3 parents $\times$ 4 constructions on the from-scratch CIFAR-10 trunks; seed 0, holdout 4.

B.2 CIFAR-100, leakage-free

space	draws	probe	top-1	eucl	mahaT	per-ev
<i>parent: SupCon</i>						
(parent)	4	0.930 $\pm$ 0.017	0.892 $\pm$ 0.003	0.563 $\pm$ 0.116	0.587 $\pm$ 0.078	0.01 $\pm$ 0.02
res (residual)	4	0.923 $\pm$ 0.023	0.860 $\pm$ 0.022	0.695 $\pm$ 0.054	0.646 $\pm$ 0.065	0.23 $\pm$ 0.21
res (concat)	4	0.957 $\pm$ 0.014	0.896 $\pm$ 0.002	0.646 $\pm$ 0.137	0.631 $\pm$ 0.063	0.12 $\pm$ 0.19
res-nplm (residual)	4	0.926 $\pm$ 0.027	0.886 $\pm$ 0.003	0.702 $\pm$ 0.067	0.712 $\pm$ 0.036	0.22 $\pm$ 0.19
res-nplm (concat)	4	0.946 $\pm$ 0.020	0.899 $\pm$ 0.002	0.636 $\pm$ 0.122	0.658 $\pm$ 0.061	0.10 $\pm$ 0.16
<i>parent: SupCon+SIGReg ($\lambda=5$)</i>						
(parent)	4	0.912 $\pm$ 0.020	0.891 $\pm$ 0.004	0.437 $\pm$ 0.080	0.416 $\pm$ 0.040	0.02 $\pm$ 0.01
res (residual)	4	0.902 $\pm$ 0.030	0.652 $\pm$ 0.011	0.396 $\pm$ 0.011	0.435 $\pm$ 0.040	0.01 $\pm$ 0.01
res (concat)	4	0.949 $\pm$ 0.014	0.891 $\pm$ 0.003	0.410 $\pm$ 0.048	0.402 $\pm$ 0.042	0.01 $\pm$ 0.01
res-nplm (residual)	4	0.921 $\pm$ 0.021	0.880 $\pm$ 0.003	0.710 $\pm$ 0.035	0.629 $\pm$ 0.030	0.16 $\pm$ 0.04
res-nplm (concat)	4	0.943 $\pm$ 0.015	0.895 $\pm$ 0.002	0.544 $\pm$ 0.070	0.534 $\pm$ 0.019	0.06 $\pm$ 0.02
<i>parent: NPLM-dist. (sup.)</i>						
(parent)	4	0.788 $\pm$ 0.079	0.815 $\pm$ 0.013	0.665 $\pm$ 0.026	0.685 $\pm$ 0.047	0.17 $\pm$ 0.06
res (residual)	4	0.884 $\pm$ 0.020	0.670 $\pm$ 0.025	0.429 $\pm$ 0.086	0.439 $\pm$ 0.085	0.03 $\pm$ 0.02
res (concat)	4	0.908 $\pm$ 0.023	0.831 $\pm$ 0.005	0.452 $\pm$ 0.089	0.562 $\pm$ 0.094	0.03 $\pm$ 0.03
res-nplm (residual)	4	0.841 $\pm$ 0.041	0.710 $\pm$ 0.056	0.480 $\pm$ 0.033	0.487 $\pm$ 0.077	0.01 $\pm$ 0.01
res-nplm (concat)	4	0.879 $\pm$ 0.032	0.822 $\pm$ 0.011	0.596 $\pm$ 0.040	0.608 $\pm$ 0.068	0.09 $\pm$ 0.07

Table 23: CIFAR-10 leakage-free lineage: the residual constructions across holdout draws. Mean $\pm$ sd over the archived holdout and the random draws; the single-holdout table above is the archived class alone. *Coverage:* from-scratch CIFAR-10 trunks, single holdout, holdouts [4, 7, 8, 9]; seed 0 per holdout; mean $\pm$ sd over holdouts.

Table 24: CIFAR-100, leakage-free lineage: every (objective, construction) $\times$ (pre, post). (A) Probe and per-event geometry. *Coverage:* 8/8 objectives, from random init, 100-D, holdout 4 (single holdout, $b = 0.01$), seed 0. ‘post’ = the exp-68 fine-tuning loop (distance pool, 2 rounds), discovery from a 2% injected sample; per-fraction post geometry present for 8/8 objectives (‘–’ otherwise; the whole-class natural pass is in the pools table). Residual concats: parents SupCon / SupCon+SIGReg / NPLM-dist. only (exp 137); discovery on scratch concats not run.

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
<i>SupCon</i>								
plain	0.940	0.931	0.233	0.304	0.279	0.352	0.000	0.010
+res concat	0.942	–	0.521	–	0.331	–	0.030	–
+res-nplm concat	0.947	–	0.227	–	0.267	–	0.000	–
<i>SupCon+SIGReg ($\lambda 5$)</i>								
plain	0.937	0.881	0.620	0.460	0.602	0.493	0.070	0.020
+res concat	0.949	–	0.648	–	0.545	–	0.080	–
+res-nplm concat	0.904	–	0.575	–	0.580	–	0.040	–
<i>NPLM-dist. +cw SIGReg</i>								
plain	0.898	0.929	0.525	0.500	0.411	0.413	0.030	0.010
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>SupCon+SIGReg (repulse)</i>								
plain	0.823	0.817	0.340	0.341	0.320	0.360	0.000	0.030
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>								
plain	0.820	0.890	0.408	0.296	0.345	0.275	0.050	0.000
+res concat	0.867	–	0.521	–	0.418	–	0.010	–
+res-nplm concat	0.849	–	0.516	–	0.462	–	0.040	–
<i>SimCLR</i>								
plain	0.799	0.888	0.443	0.310	0.385	0.281	0.010	0.020
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>VISReg/LeJEPa</i>								
plain	0.797	0.896	0.499	0.288	0.382	0.264	0.030	0.010
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>NPLM (unsup.)</i>								
plain	0.757	0.869	0.413	0.414	0.361	0.271	0.010	0.050
+res concat	–	–	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–	–	–

Table 25: CIFAR-100, leakage-free lineage: every (objective, construction) $\times$ (pre, post). (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage*: 8/8 objectives, from random init, 100-D, holdout 4 (single holdout, $b = 0.01$), seed 0. ‘post’ = the exp-68 fine-tuning loop (distance pool, 2 rounds), discovery from a 2% injected sample; per-fraction post geometry present for 8/8 objectives (‘-’ otherwise; the whole-class natural pass is in the pools table). Residual concats: parents SupCon / SupCon+SIGReg / NPLM-dist. only (exp 137); discovery on scratch concats not run.

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
<i>SupCon</i>						
plain	0.14	0.06	0.04	0.02	0.50	0.88
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>SupCon+SIGReg ($\lambda 5$)</i>						
plain	0.22	0.18	0.06	0.06	0.70	0.92
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>NPLM-dist.+cw SIGReg</i>						
plain	0.20	0.20	0.06	0.04	0.58	0.56
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>SupCon+SIGReg (repulse)</i>						
plain	0.14	0.08	0.02	0.02	0.38	0.52
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>						
plain	0.18	0.06	0.04	0.00	0.52	0.78
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>SimCLR</i>						
plain	0.06	0.00	0.02	0.00	0.18	0.70
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>VISReg/LeJEPa</i>						
plain	0.04	0.12	0.06	0.00	0.08	0.60
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>NPLM (unsup.)</i>						
plain	0.00	0.02	0.02	0.00	0.12	0.46
+res concat	–	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–

objective	probe	top-1	acc	supAUC	eucl	mahaT	mahaPC	lid	per-ev	probe sd
SupCon	0.940	0.608	0.545	0.974	0.233	0.279	0.487	0.740	0.00	0.001
SupCon+SIGReg ($\lambda=5$)	0.937	0.597	0.588	0.976	0.620	0.602	0.631	0.734	0.07	0.007
NPLM-dist.+classwise SIGReg	0.898	0.507	0.463	0.946	0.525	0.411	0.553	0.678	0.03	0.000
SupCon+SIGReg (repulse)	0.823	0.360	0.346	0.930	0.340	0.320	0.432	0.537	0.00	0.008
NPLM-dist. (sup.)	0.820	0.254	0.232	0.913	0.408	0.345	0.348	0.542	0.05	0.003
SimCLR	0.799	0.292	0.242	0.859	0.443	0.385	0.453	0.570	0.01	0.012
VISReg/LeJEPa	0.797	0.239	0.190	0.844	0.499	0.382	0.434	0.534	0.03	0.004
NPLM (unsup.)	0.757	0.223	0.181	0.836	0.413	0.361	0.447	0.635	0.01	0.003

Table 26: CIFAR-100 leakage-free lineage: the frozen-space battery. probe = holdout-vs-rest linear AUC; top-1 = supervised linear probe on seen classes; acc = nearest-centroid; supAUC = macro OvR AUC of the proto posterior; eucl / mahaT / mahaPC = novelty AUC of min-centroid distance, tied and per-class Mahalanobis; lid = local-intrinsic-dimension AUC; per-ev = per-event power at $\alpha=0.05$. *Coverage:* 8/8 objectives; CIFAR-100 from random init, 100-D, holdout class 4 (single holdout, $b = 0.01$), seed 0; probe and top-1 over 3 probe restarts.

objective	SparKer			Mahalanobis			MMD		
	0.01	0.02	0.05	0.01	0.02	0.05	0.01	0.02	0.05
SupCon	0.08	0.14	1.00	0.08	0.04	0.00	0.28	0.50	1.00
SupCon+SIGReg ($\lambda=5$)	0.20	0.22	1.00	0.10	0.06	0.48	0.30	0.70	1.00
NPLM-dist.+classwise SIGReg	0.06	0.20	1.00	0.08	0.06	0.08	0.24	0.58	1.00
SupCon+SIGReg (repulse)	0.08	0.14	1.00	0.02	0.02	0.00	0.20	0.38	1.00
NPLM-dist. (sup.)	0.08	0.18	0.90	0.02	0.04	0.00	0.20	0.52	1.00
SimCLR	0.02	0.06	1.00	0.04	0.02	0.02	0.08	0.18	0.82
VISReg/LeJEPA	0.02	0.04	0.86	0.04	0.06	0.00	0.06	0.08	0.68
NPLM (unsup.)	0.04	0.00	1.00	0.02	0.02	0.00	0.08	0.12	0.94

Table 27: CIFAR-100 leakage-free lineage: dataset-level detection power on the frozen space, before any discovery. Signal fraction f of held-out images injected into a seen-class test sample. *Coverage*: 8/8 objectives; annealed- σ SparKer, parametric Mahalanobis and MMD two-sample power at $\alpha=0.05$, $N_D=5000$, 200 null / 50 signal toys; fractions 0.01, 0.02, 0.05.

objective	frozen pools				legal cut		loop, 2% injected			loop, whole class		
	dist r1	np r1	np r2	np+BN r2	$n_{\min}=10$	$n_{\min}=30$	purity	probe	mahaT	purity	probe	pre→post
SupCon	0.000	0.052	0.021	0.027	ref.	ref.	0.000	0.931	0.352	0.000	0.939→0.933	
SupCon+SIGReg ($\lambda=5$)	0.008	0.007	0.018	0.018	ref.	ref.	0.008	0.881	0.493	0.008	0.941→0.904	
NPLM-dist.+classwise SIGReg	0.006	0.031	0.020	0.016	0.048	ref.	0.006	0.929	0.413	0.006	0.897→0.934	
SupCon+SIGReg (repulse)	0.009	0.024	0.037	0.040	ref.	ref.	0.009	0.817	0.360	0.009	0.842→0.857	
NPLM-dist. (sup.)	0.008	0.004	0.001	0.020	ref.	ref.	0.008	0.890	0.275	0.008	0.817→0.851	
SimCLR	0.004	0.024	0.014	0.005	ref.	ref.	0.004	0.888	0.281	0.004	0.818→0.888	
VISReg/LeJEPa	0.005	0.022	0.015	0.018	ref.	ref.	0.005	0.896	0.264	0.005	0.813→0.898	
NPLM (unsup.)	0.009	0.022	0.014	0.016	ref.	ref.	0.009	0.869	0.271	0.009	0.761→0.879	

Table 28: CIFAR-100 leakage-free lineage: pools and discovery. Round-1 purity of the frozen distance and density-ratio (np) pools, the np pool’s round 2 with BN frozen and adapting, the label-free cut, then the fine-tuning loop for *discovery from a 2% injected sample* of the held-out class (the paper’s regime): round-1 purity, post probe and post tied Mahalanobis. The last two columns are the loop’s natural pass, in which the whole held-out class is present in the unlabelled bank — a different, easier regime, shown for reference only. *Coverage:* 8/8 objectives; frozen pools = trunk frozen (BN in eval unless ‘BN adapt’), 2×2 -epoch anchor rounds; legal cut at $n_{\min} = 10$ and 30 (‘ref.’ = the rule declined); loop = exp-68 fine-tuning loop, distance pool, 2×5 epochs. Injected-sample probe/mahaT present for 8/8 objectives (‘-’ = npz predates the per-fraction keys).

objective	draws	pre-discovery				frozen	loop		
		probe	top-1	mahaT	per-ev	purity	purity 2%	purity whole	probe post whole
SupCon	4	0.955±0.020	0.604±0.005	0.447±0.117	0.033±0.034	0.021±0.018	0.005±0.006	0.005±0.006	0.957±0.021
SupCon+SIGReg ($\lambda=5$)	4	0.947±0.019	0.596±0.005	0.549±0.054	0.048±0.018	0.032±0.021	0.009±0.003	0.008±0.003	0.933±0.022
NPLM-dist. (sup.)	4	0.857±0.045	0.255±0.003	0.484±0.161	0.068±0.048	0.030±0.031	0.016±0.012	0.016±0.013	0.896±0.044
NPLM-dist.+classwise SIGReg	4	0.944±0.031	0.510±0.002	0.539±0.129	0.122±0.128	0.034±0.026	0.025±0.022	0.025±0.022	0.949±0.023

Table 29: CIFAR-100 from scratch, across holdout draws. Pre-discovery probe, top-1, tied Mahalanobis and per-event power; frozen density-ratio pool purity; the fine-tuning loop’s round-1 purity for discovery from a 2% injected sample and, for reference, with the whole held-out class present in the bank, with the whole-class post probe. *Coverage:* CIFAR-100 from scratch, single holdout, holdouts [4, 43, 48, 57] (the archived class plus random draws), one seed per draw; objectives with fewer than two draws omitted; injected-sample purity from the POST-grid logs (‘-’ = run predates it).

space	probe	top-1	acc	eucl	mahaT	lid	per-ev
<i>parent: SupCon</i>							
(parent)	0.940	0.608	0.545	0.233	0.279	0.740	0.00
res (residual)	0.869	0.344	0.248	0.607	0.474	0.630	0.05
res (concat)	0.942	0.604	0.479	0.521	0.331	0.650	0.03
res-nplm (residual)	0.882	0.443	0.346	0.291	0.355	0.638	0.00
res-nplm (concat)	0.947	0.602	0.550	0.227	0.267	0.731	0.00
<i>parent: SupCon+SIGReg ($\lambda=5$)</i>							
(parent)	0.937	0.597	0.588	0.620	0.602	0.734	0.07
res (residual)	0.860	0.248	0.178	0.604	0.473	0.597	0.08
res (concat)	0.949	0.595	0.575	0.648	0.545	0.729	0.08
res-nplm (residual)	0.900	0.575	0.531	0.464	0.484	0.716	0.01
res-nplm (concat)	0.904	0.595	0.591	0.575	0.580	0.739	0.04
<i>parent: NPLM-dist. (sup.)</i>							
(parent)	0.820	0.254	0.232	0.408	0.345	0.542	0.05
res (residual)	0.834	0.266	0.221	0.524	0.449	0.567	0.01
res (concat)	0.867	0.365	0.236	0.521	0.418	0.562	0.01
res-nplm (residual)	0.827	0.316	0.283	0.527	0.480	0.574	0.06
res-nplm (concat)	0.849	0.391	0.349	0.516	0.462	0.588	0.04

Table 30: CIFAR-100 leakage-free lineage: every residual construction, with the supervised top-1 that the main text’s no-cost claim rests on. *Coverage:* 3 parents $\times$ 4 constructions on the from-scratch CIFAR-100 trunks; seed 0, holdout 4.

B.3 CIFAR-10 and CIFAR-100, hub-pretrained (ablation only)

The hub-pretrained CIFAR cells — every number in our earlier technical report — use an encoder that saw the held-out class. We keep them as an ablation, and the grids below make their coverage explicit: standalone arms (exps 50/53) have pre-discovery geometry and power, and post-discovery probe and power from the proto fine-tune (exp 55) or the NPLM fine-tune (exp 58), but their post-discovery *geometry* was never recorded; the 16+16 concats (exp 59) and the residual fine-tunes (exps 73/74) carry both. The multi-seed configurations are Tables 10 and 11, the five-seed residual concats Table 12. Their probe records (0.983 ± 0.001 post-discovery on CIFAR-10; 0.951 ± 0.005 on CIFAR-100) reproduce across seeds; their leakage-free counterparts are 0.954 (NPLM-dist.+classwise SIGReg under the loop, whole-class pass) and 0.949 (SupCon+SIGReg residual concat).

space	draws	probe	top-1	eucl	mahaT	per-ev
<i>parent: SupCon</i>						
(parent)	4	0.955±0.020	0.604±0.005	0.463±0.170	0.447±0.117	0.03±0.03
res (residual)	4	0.925±0.036	0.341±0.009	0.538±0.132	0.473±0.119	0.05±0.03
res (concat)	4	0.963±0.018	0.598±0.004	0.535±0.066	0.457±0.077	0.05±0.03
res-nplm (residual)	4	0.929±0.036	0.477±0.023	0.483±0.216	0.464±0.120	0.10±0.16
res-nplm (concat)	4	0.963±0.018	0.599±0.005	0.484±0.219	0.446±0.135	0.07±0.11
<i>parent: SupCon+SIGReg ($\lambda=5$)</i>						
(parent)	4	0.947±0.019	0.596±0.005	0.569±0.036	0.549±0.054	0.05±0.02
res (residual)	4	0.906±0.033	0.245±0.006	0.572±0.092	0.465±0.075	0.07±0.06
res (concat)	4	0.959±0.019	0.591±0.005	0.582±0.067	0.499±0.070	0.05±0.03
res-nplm (residual)	4	0.932±0.024	0.567±0.005	0.535±0.137	0.529±0.116	0.05±0.04
res-nplm (concat)	4	0.941±0.028	0.590±0.004	0.573±0.044	0.571±0.013	0.05±0.01
<i>parent: NPLM-dist. (sup.)</i>						
(parent)	4	0.857±0.045	0.255±0.003	0.536±0.131	0.484±0.161	0.07±0.05
res (residual)	4	0.892±0.042	0.251±0.015	0.619±0.107	0.580±0.116	0.11±0.09
res (concat)	4	0.917±0.037	0.358±0.008	0.622±0.108	0.569±0.112	0.10±0.09
res-nplm (residual)	4	0.905±0.054	0.316±0.014	0.582±0.040	0.560±0.075	0.08±0.03
res-nplm (concat)	4	0.924±0.050	0.390±0.011	0.581±0.049	0.557±0.098	0.07±0.04

Table 31: CIFAR-100 leakage-free lineage: the residual constructions across holdout draws. Mean $\pm$ sd over the archived holdout and the random draws; the single-holdout table above is the archived class alone. *Coverage:* from-scratch CIFAR-100 trunks, single holdout, holdouts [4, 43, 48, 57]; seed 0 per holdout; mean $\pm$ sd over holdouts.

Table 32: CIFAR-10, hub-pretrained lineage (ablation): every (space, construction) $\times$ (pre, post). (A) Probe and per-event geometry. *Coverage*: 11 standalone arms; hub-pretrained trunk (the encoder saw the held-out class – ablation lineage); holdout 4, seed 0; post geometry for standalone arms was never measured (exp 55 recorded probe and power only).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
<i>Standalone objectives, 32-D (exps 50/53; post = exp 55 proto fine-tune)</i>								
SupCon	0.942	–	0.689	–	0.644	–	0.000	–
SupCon+SIGReg ($\lambda 1$)	0.882	–	0.637	–	0.577	–	0.000	–
NPLM-dist. (sup.)	0.783	0.963	0.726	–	0.723	–	0.200	0.346
same, NPLM fine-tune (exp 58)	–	0.812	–	–	–	–	–	0.572
NPLM-dist.+cw SIGReg	0.872	0.977	0.752	–	0.729	–	0.163	0.316
same, NPLM fine-tune (exp 58)	–	0.836	–	–	–	–	–	0.575
NPLM-bilinear	0.853	0.892	0.510	–	0.494	–	0.057	0.073
NPLM-bil.+cw SIGReg	0.854	–	0.570	–	0.564	–	0.066	–
NPLM-bil.(sup.)+cw SIGReg	0.781	–	0.516	–	0.514	–	0.045	–
NPLM-dist. (unsup.)	0.790	–	0.500	–	0.481	–	0.005	–
SimCLR	0.867	–	0.236	–	0.229	–	0.000	–
SimCLR+SIGReg	0.851	–	0.311	–	0.285	–	0.002	–
LeJEPa	0.863	–	0.408	–	0.371	–	0.007	–
<i>Concatenated / residual spaces, 16+16 (exp 59; post = discovery on the concat)</i>								
sup→res-nplm	0.934	0.984	0.834	–	0.768	–	0.275	0.511
nplmsup→res	0.865	0.896	0.432	–	0.596	–	0.015	0.076
supcon+nplm	0.954	0.920	0.768	–	0.741	–	0.015	0.116
sup+nplm	0.932	0.979	0.791	–	0.771	–	0.232	0.438
nplmsup+nplm	0.887	0.918	0.705	–	0.760	–	0.138	0.214
<i>Residual fine-tune, 32+32 (exps 73/74/80)</i>								
SupCon parent	0.940	–	0.717	–	0.631	–	0.000	–
+res concat	0.954	0.953	0.646	0.722	0.548	0.621	0.039	0.320
+res-nplm concat	0.947	0.953	0.703	0.794	0.565	0.622	0.000	0.167
res child alone	0.859	–	0.473	–	0.466	–	0.026	–
res-nplm child alone	0.895	–	0.542	–	0.531	–	0.002	–
SupCon+SIGReg ($\lambda 1$) parent	0.856	–	0.692	–	0.528	–	0.000	–
+res concat	0.914	–	0.555	–	0.426	–	0.011	–

Table 33: CIFAR-10, hub-pretrained lineage (ablation): every (space, construction) $\times$ (pre, post). (B)
Dataset-level power at $f = 0.02$, $\alpha = 0.05$. Coverage: 11 standalone arms; hub-pretrained trunk (the encoder saw the held-out class – ablation lineage); holdout 4, seed 0; post geometry for standalone arms was never measured (exp 55 recorded probe and power only).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
<i>Standalone objectives, 32-D (exps 50/53; post = exp 55 proto fine-tune)</i>						
SupCon	0.60	–	0.28	–	0.88	–
SupCon+SIGReg ($\lambda 1$)	0.64	–	0.10	–	0.74	–
NPLM-dist. (sup.)	0.86	0.04	0.46	0.50	0.28	0.40
same, NPLM fine-tune (exp 58)	–	1.00	–	0.90	–	0.74
NPLM-dist.+cw SIGReg	0.70	0.06	0.32	0.74	0.12	0.34
same, NPLM fine-tune (exp 58)	–	0.92	–	0.56	–	0.72
NPLM-bilinear	0.10	0.04	0.16	0.36	0.06	0.00
NPLM-bil.+cw SIGReg	0.10	–	0.26	–	0.06	–
NPLM-bil.(sup.)+cw SIGReg	0.04	–	0.08	–	0.00	–
NPLM-dist. (unsup.)	0.26	–	0.14	–	0.00	–
SimCLR	0.06	–	0.00	–	0.36	–
SimCLR+SIGReg	0.06	–	0.00	–	0.14	–
LeJEPA	0.16	–	0.10	–	0.04	–
<i>Concatenated / residual spaces, 16+16 (exp 59; post = discovery on the concat)</i>						
sup $\rightarrow$ res-nplm	0.88	1.00	0.70	0.90	0.34	0.90
nplmsup $\rightarrow$ res	0.04	0.08	0.30	0.36	0.00	0.02
supcon+nplm	0.26	0.30	0.64	0.88	0.66	0.06
sup+nplm	0.42	0.94	0.84	0.86	0.24	0.80
nplmsup+nplm	0.12	0.04	0.66	0.62	0.08	0.10
<i>Residual fine-tune, 32+32 (exps 73/74/80)</i>						
SupCon parent	0.98	–	–	–	–	–
+res concat	0.26	0.22	–	0.46	–	0.18
+res-nplm concat	0.92	0.58	–	0.80	–	0.42
res child alone	0.00	–	–	–	–	–
res-nplm child alone	0.06	–	–	–	–	–
SupCon+SIGReg ($\lambda 1$) parent	–	–	–	–	–	–
+res concat	–	–	–	–	–	–

Table 34: CIFAR-100, hub-pretrained lineage (ablation): every (space, construction) $\times$ (pre, post). (A) Probe and per-event geometry. *Coverage*: 11 standalone arms; hub-pretrained trunk (the encoder saw the held-out class – ablation lineage); holdout 4, seed 0; post geometry for standalone arms was never measured (exp 55 recorded probe and power only).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
<i>Standalone objectives, 32-D (exps 50/53; post = exp 55 proto fine-tune)</i>								
SupCon	0.942	–	0.332	–	0.341	–	0.000	–
SupCon+SIGReg ($\lambda 1$)	0.883	–	0.436	–	0.443	–	0.000	–
NPLM-dist. (sup.)	0.803	0.837	0.369	–	0.269	–	0.020	0.010
NPLM-dist.+cw SIGReg	0.886	0.886	0.452	–	0.372	–	0.020	0.020
NPLM-bilinear	0.935	0.891	0.532	–	0.484	–	0.070	0.050
NPLM-bil.+cw SIGReg	0.872	–	0.458	–	0.388	–	0.030	–
NPLM-bil.(sup.)+cw SIGReg	0.925	–	0.377	–	0.316	–	0.010	–
NPLM-dist. (unsup.)	0.737	–	0.477	–	0.409	–	0.010	–
SimCLR	0.854	–	0.395	–	0.336	–	0.010	–
SimCLR+SIGReg	0.863	–	0.519	–	0.458	–	0.040	–
LeJEPa	0.868	–	0.473	–	0.403	–	0.010	–
<i>Concatenated / residual spaces, 16+16 (exp 59; post = discovery on the concat)</i>								
sup $\rightarrow$ res-nplm	0.798	0.800	0.490	–	0.325	–	0.000	0.020
nplmsup $\rightarrow$ res	0.845	0.866	0.516	–	0.417	–	0.010	0.000
supcon+nplm	0.942	0.932	0.436	–	0.415	–	0.000	0.000
sup+nplm	0.896	0.860	0.567	–	0.417	–	0.020	0.090
nplmsup+nplm	0.896	0.900	0.526	–	0.489	–	0.030	0.000
<i>Residual fine-tune, 100+100 (exps 73/74/80)</i>								
SupCon parent	0.958	–	0.338	–	0.347	–	0.000	–
+res concat	0.959	0.957	0.575	0.562	0.440	0.483	0.020	0.030
+res-nplm concat	0.957	0.950	0.296	0.530	0.312	0.455	0.000	0.020
res child alone	0.839	–	0.650	–	0.558	–	0.080	–
res-nplm child alone	0.910	–	0.317	–	0.318	–	0.000	–
SupCon+SIGReg ($\lambda 1$) parent	0.945	–	0.388	–	0.381	–	0.000	–
+res concat	0.942	–	0.568	–	0.479	–	0.040	–

Table 35: CIFAR-100, hub-pretrained lineage (ablation): every (space, construction) $\times$ (pre, post). (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage*: 11 standalone arms; hub-pretrained trunk (the encoder saw the held-out class – ablation lineage); holdout 4, seed 0; post geometry for standalone arms was never measured (exp 55 recorded probe and power only).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
<i>Standalone objectives, 32-D (exps 50/53; post = exp 55 proto fine-tune)</i>						
SupCon	0.18	–	0.06	–	0.64	–
SupCon+SIGReg ($\lambda 1$)	0.02	–	0.06	–	0.48	–
NPLM-dist. (sup.)	0.14	0.18	0.00	0.02	0.78	0.68
NPLM-dist.+cw SIGReg	0.04	0.12	0.02	0.04	0.52	0.66
NPLM-bilinear	0.04	0.04	0.04	0.08	0.30	0.58
NPLM-bil.+cw SIGReg	0.06	–	0.00	–	0.22	–
NPLM-bil.(sup.)+cw SIGReg	0.06	–	0.00	–	0.68	–
NPLM-dist. (unsup.)	0.08	–	0.04	–	0.24	–
SimCLR	0.12	–	0.00	–	0.14	–
SimCLR+SIGReg	0.00	–	0.00	–	0.10	–
LeJEPA	0.04	–	0.10	–	0.16	–
<i>Concatenated / residual spaces, 16+16 (exp 59; post = discovery on the concat)</i>						
sup $\rightarrow$ res-nplm	0.26	0.18	0.04	0.00	0.30	0.52
nplmsup $\rightarrow$ res	0.04	0.22	0.04	0.00	0.12	0.52
supcon+nplm	0.26	0.56	0.00	0.08	0.48	0.90
sup+nplm	0.10	0.06	0.04	0.08	0.30	0.46
nplmsup+nplm	0.16	0.14	0.02	0.00	0.62	0.62
<i>Residual fine-tune, 100+100 (exps 73/74/80)</i>						
SupCon parent	0.20	–	–	–	–	–
+res concat	0.18	0.06	–	0.08	–	0.32
+res-nplm concat	0.30	0.18	–	0.02	–	0.84
res child alone	0.24	–	–	–	–	–
res-nplm child alone	0.08	–	–	–	–	–
SupCon+SIGReg ($\lambda 1$) parent	–	–	–	–	–	–
+res concat	–	–	–	–	–	–

C galaxy10: every measurement, across holdout draws

Galaxy10 is the only dataset genuinely out of distribution for all three backbones, and it carries the most weight. Every cell is single-holdout (one morphology of ten, $b \approx 0.1$) and resampled over five held-out classes $\{2, 3, 4, 5, 6\}$; the archived class-9 draw is reported only as a reference. Table 42 is the fine-tuning battery on all three backbones; Table 43 the residual constructions on the same draws; Table 4 the frozen density-ratio pool over draws and seeds; Table 44 the supervised top-1 that fixes the cost of SIGReg on this dataset.

The summary grids below give every (objective, construction) $\times$ (pre, post) cell per backbone as mean $\pm$ sd across the five draws. The residual concats are built on the SupCon and SupCon+SIGReg parents; discovery on a residual concat (exp 72) was run on the SupCon+res concat only, so the other concats’ post columns are dashes. The after-discovery construction of Section 5 (exp 134c) is a different pipeline order and is tabulated in Table 14.

C.1 galaxy10 / DINO

Table 36: Galaxy10 / DINO: every (objective, construction) $\times$ (pre, post), across holdout draws. (A) Probe and per-event geometry. *Coverage:* Galaxy10 / DINO, single holdout, 5 draws (classes 2,3,4,5,6), mean $\pm$ sd across draws, one seed per draw; plain-arm post columns = discovery from a 2% injected sample (per-fraction post geometry present in 5/5 draws, ‘–’ otherwise; the whole-class natural pass is in Table 42); residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
<i>SupCon</i>								
plain	0.879 $\pm$ 0.026	0.860 $\pm$ 0.032	0.526 $\pm$ 0.145	0.548 $\pm$ 0.150	0.500 $\pm$ 0.135	0.488 $\pm$ 0.141	0.063 $\pm$ 0.092	0.086 $\pm$ 0.088
+res concat	0.903 $\pm$ 0.027	0.886 $\pm$ 0.024	0.497 $\pm$ 0.122	0.506 $\pm$ 0.135	0.488 $\pm$ 0.138	0.470 $\pm$ 0.146	0.059 $\pm$ 0.078	0.063 $\pm$ 0.077
+res-nplm concat	0.890 $\pm$ 0.023	–	0.538 $\pm$ 0.140	–	0.494 $\pm$ 0.134	–	0.062 $\pm$ 0.088	–
<i>SupCon+SIGReg (λ5)</i>								
plain	0.804 $\pm$ 0.064	0.849 $\pm$ 0.033	0.499 $\pm$ 0.104	0.509 $\pm$ 0.094	0.482 $\pm$ 0.114	0.478 $\pm$ 0.097	0.040 $\pm$ 0.038	0.040 $\pm$ 0.029
+res concat	0.869 $\pm$ 0.026	–	0.494 $\pm$ 0.089	–	0.486 $\pm$ 0.116	–	0.052 $\pm$ 0.040	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>								
plain	0.775 $\pm$ 0.097	0.782 $\pm$ 0.086	0.585 $\pm$ 0.123	0.536 $\pm$ 0.049	0.556 $\pm$ 0.129	0.565 $\pm$ 0.092	0.105 $\pm$ 0.121	0.080 $\pm$ 0.053
<i>SimCLR</i>								
plain	0.790 $\pm$ 0.064	0.810 $\pm$ 0.087	0.460 $\pm$ 0.125	0.497 $\pm$ 0.108	0.476 $\pm$ 0.111	0.481 $\pm$ 0.111	0.049 $\pm$ 0.046	0.045 $\pm$ 0.043
<i>SimCLR+SIGReg</i>								
plain	0.801 $\pm$ 0.084	0.846 $\pm$ 0.036	0.477 $\pm$ 0.069	0.502 $\pm$ 0.078	0.508 $\pm$ 0.093	0.496 $\pm$ 0.070	0.045 $\pm$ 0.029	0.039 $\pm$ 0.023
<i>NPLM-bilinear</i>								
plain	0.684 $\pm$ 0.134	0.811 $\pm$ 0.032	0.483 $\pm$ 0.063	0.491 $\pm$ 0.070	0.450 $\pm$ 0.120	0.462 $\pm$ 0.125	0.045 $\pm$ 0.036	0.066 $\pm$ 0.084

Table 37: Galaxy10 / DINO: every (objective, construction) $\times$ (pre, post), across holdout draws. (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* Galaxy10 / DINO, single holdout, 5 draws (classes 2,3,4,5,6), mean $\pm$ sd across draws, one seed per draw; plain-arm post columns = discovery from a 2% injected sample (per-fraction post geometry present in 5/5 draws, ‘–’ otherwise; the whole-class natural pass is in Table 42); residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
<i>SupCon</i>						
plain	0.04 $\pm$ 0.03	0.10 $\pm$ 0.05	0.06 $\pm$ 0.04	0.11 $\pm$ 0.06	0.08 $\pm$ 0.08	0.10 $\pm$ 0.07
+res concat	0.04	0.05 $\pm$ 0.03	–	0.10 $\pm$ 0.06	–	0.13 $\pm$ 0.13
+res-nplm concat	0.04	–	–	–	–	–
<i>SupCon+SIGReg (λ_5)</i>						
plain	0.06 $\pm$ 0.03	0.10 $\pm$ 0.05	0.06 $\pm$ 0.04	0.10 $\pm$ 0.07	0.09 $\pm$ 0.06	0.15 $\pm$ 0.11
+res concat	0.02	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>						
plain	0.05 $\pm$ 0.05	0.07 $\pm$ 0.05	0.06 $\pm$ 0.03	0.08 $\pm$ 0.05	0.10 $\pm$ 0.08	0.12 $\pm$ 0.11
<i>SimCLR</i>						
plain	0.05 $\pm$ 0.03	0.06 $\pm$ 0.05	0.05 $\pm$ 0.04	0.08 $\pm$ 0.06	0.08 $\pm$ 0.06	0.08 $\pm$ 0.09
<i>SimCLR+SIGReg</i>						
plain	0.04 $\pm$ 0.02	0.04 $\pm$ 0.01	0.08 $\pm$ 0.06	0.11 $\pm$ 0.09	0.09 $\pm$ 0.05	0.10 $\pm$ 0.10
<i>NPLM-bilinear</i>						
plain	0.10 $\pm$ 0.04	0.08 $\pm$ 0.06	0.05 $\pm$ 0.05	0.08 $\pm$ 0.04	0.05 $\pm$ 0.05	0.09 $\pm$ 0.08

C.2 galaxy10 / LeJEPa

Table 38: Galaxy10 / LeJEPa: every (objective, construction) $\times$ (pre, post), across holdout draws. (A) Probe and per-event geometry. *Coverage:* Galaxy10 / LeJEPa, single holdout, 5 draws (classes 2,3,4,5,6), mean $\pm$ sd across draws, one seed per draw; plain-arm post columns = discovery from a 2% injected sample (per-fraction post geometry present in 5/5 draws, ‘–’ otherwise; the whole-class natural pass is in Table 42); residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
<i>SupCon</i>								
plain	0.892 $\pm$ 0.056	0.912 $\pm$ 0.053	0.591 $\pm$ 0.080	0.683 $\pm$ 0.146	0.711 $\pm$ 0.115	0.713 $\pm$ 0.127	0.047 $\pm$ 0.027	0.048 $\pm$ 0.040
+res concat	0.945 $\pm$ 0.033	0.936 $\pm$ 0.034	0.630 $\pm$ 0.059	0.628 $\pm$ 0.084	0.709 $\pm$ 0.079	0.743 $\pm$ 0.085	0.124 $\pm$ 0.059	0.155 $\pm$ 0.108
+res-nplm concat	0.931 $\pm$ 0.045	–	0.644 $\pm$ 0.080	–	0.699 $\pm$ 0.089	–	0.086 $\pm$ 0.061	–
<i>SupCon+SIGReg (λ5)</i>								
plain	0.762 $\pm$ 0.181	0.827 $\pm$ 0.118	0.633 $\pm$ 0.148	0.650 $\pm$ 0.093	0.665 $\pm$ 0.163	0.670 $\pm$ 0.124	0.134 $\pm$ 0.119	0.144 $\pm$ 0.110
+res concat	0.855 $\pm$ 0.094	–	0.627 $\pm$ 0.068	–	0.692 $\pm$ 0.118	–	0.093 $\pm$ 0.047	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>								
plain	0.787 $\pm$ 0.073	0.798 $\pm$ 0.125	0.591 $\pm$ 0.133	0.589 $\pm$ 0.153	0.613 $\pm$ 0.174	0.625 $\pm$ 0.130	0.132 $\pm$ 0.095	0.033 $\pm$ 0.021
<i>SimCLR</i>								
plain	0.829 $\pm$ 0.066	0.847 $\pm$ 0.067	0.488 $\pm$ 0.086	0.561 $\pm$ 0.056	0.574 $\pm$ 0.107	0.614 $\pm$ 0.047	0.050 $\pm$ 0.039	0.097 $\pm$ 0.125
<i>SimCLR+SIGReg</i>								
plain	0.836 $\pm$ 0.060	0.846 $\pm$ 0.055	0.519 $\pm$ 0.134	0.511 $\pm$ 0.119	0.596 $\pm$ 0.118	0.563 $\pm$ 0.101	0.065 $\pm$ 0.070	0.102 $\pm$ 0.125
<i>NPLM-bilinear</i>								
plain	0.808 $\pm$ 0.091	0.835 $\pm$ 0.079	0.581 $\pm$ 0.065	0.571 $\pm$ 0.049	0.611 $\pm$ 0.097	0.598 $\pm$ 0.087	0.075 $\pm$ 0.028	0.063 $\pm$ 0.023

Table 39: Galaxy10 / LeJEPa: every (objective, construction) $\times$ (pre, post), across holdout draws. (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* Galaxy10 / LeJEPa, single holdout, 5 draws (classes 2,3,4,5,6), mean $\pm$ sd across draws, one seed per draw; plain-arm post columns = discovery from a 2% injected sample (per-fraction post geometry present in 5/5 draws, ‘–’ otherwise; the whole-class natural pass is in Table 42); residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
<i>SupCon</i>						
plain	0.20 $\pm$ 0.22	0.24 $\pm$ 0.25	0.13 $\pm$ 0.06	0.19 $\pm$ 0.20	0.23 $\pm$ 0.04	0.26 $\pm$ 0.24
+res concat	0.04	0.11 $\pm$ 0.11	–	0.21 $\pm$ 0.18	–	0.15 $\pm$ 0.16
+res-nplm concat	0.14	–	–	–	–	–
<i>SupCon+SIGReg (λ5)</i>						
plain	0.18 $\pm$ 0.25	0.21 $\pm$ 0.18	0.15 $\pm$ 0.04	0.16 $\pm$ 0.12	0.16 $\pm$ 0.14	0.22 $\pm$ 0.19
+res concat	0.14	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>						
plain	0.18 $\pm$ 0.12	0.32 $\pm$ 0.16	0.08 $\pm$ 0.10	0.21 $\pm$ 0.18	0.16 $\pm$ 0.08	0.22 $\pm$ 0.12
<i>SimCLR</i>						
plain	0.04 $\pm$ 0.02	0.07 $\pm$ 0.05	0.14 $\pm$ 0.06	0.19 $\pm$ 0.18	0.04 $\pm$ 0.03	0.13 $\pm$ 0.09
<i>SimCLR+SIGReg</i>						
plain	0.05 $\pm$ 0.02	0.08 $\pm$ 0.06	0.08 $\pm$ 0.05	0.16 $\pm$ 0.11	0.10 $\pm$ 0.06	0.09 $\pm$ 0.07
<i>NPLM-bilinear</i>						
plain	0.08 $\pm$ 0.03	0.08 $\pm$ 0.06	0.11 $\pm$ 0.12	0.20 $\pm$ 0.15	0.07 $\pm$ 0.04	0.07 $\pm$ 0.07

C.3 galaxy10 / VISReg

Table 40: Galaxy10 / VISReg: every (objective, construction) $\times$ (pre, post), across holdout draws. (A) Probe and per-event geometry. *Coverage:* Galaxy10 / VISReg, single holdout, 5 draws (classes 2,3,4,5,6), mean $\pm$ sd across draws, one seed per draw; plain-arm post columns = discovery from a 2% injected sample (per-fraction post geometry present in 5/5 draws, ‘–’ otherwise; the whole-class natural pass is in Table 42); residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	probe		eucl		mahaT		per-event	
	pre	post	pre	post	pre	post	pre	post
<i>SupCon</i>								
plain	0.906 $\pm$ 0.043	0.905 $\pm$ 0.041	0.700 $\pm$ 0.076	0.683 $\pm$ 0.069	0.686 $\pm$ 0.074	0.672 $\pm$ 0.089	0.095 $\pm$ 0.054	0.105 $\pm$ 0.097
+res concat	0.947 $\pm$ 0.025	0.922 $\pm$ 0.037	0.647 $\pm$ 0.065	0.584 $\pm$ 0.090	0.658 $\pm$ 0.061	0.602 $\pm$ 0.046	0.087 $\pm$ 0.054	0.045 $\pm$ 0.051
+res-nplm concat	0.923 $\pm$ 0.036	–	0.709 $\pm$ 0.097	–	0.711 $\pm$ 0.106	–	0.116 $\pm$ 0.088	–
<i>SupCon+SIGReg (λ5)</i>								
plain	0.823 $\pm$ 0.108	0.907 $\pm$ 0.033	0.615 $\pm$ 0.118	0.621 $\pm$ 0.122	0.581 $\pm$ 0.136	0.591 $\pm$ 0.130	0.090 $\pm$ 0.096	0.139 $\pm$ 0.157
+res concat	0.888 $\pm$ 0.052	–	0.576 $\pm$ 0.095	–	0.586 $\pm$ 0.136	–	0.086 $\pm$ 0.062	–
+res-nplm concat	–	–	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>								
plain	0.738 $\pm$ 0.105	0.733 $\pm$ 0.186	0.640 $\pm$ 0.112	0.592 $\pm$ 0.066	0.628 $\pm$ 0.138	0.648 $\pm$ 0.112	0.129 $\pm$ 0.098	0.033 $\pm$ 0.019
<i>SimCLR</i>								
plain	0.847 $\pm$ 0.053	0.838 $\pm$ 0.081	0.486 $\pm$ 0.076	0.542 $\pm$ 0.067	0.526 $\pm$ 0.065	0.546 $\pm$ 0.062	0.045 $\pm$ 0.019	0.092 $\pm$ 0.093
<i>SimCLR+SIGReg</i>								
plain	0.835 $\pm$ 0.040	0.861 $\pm$ 0.033	0.515 $\pm$ 0.084	0.492 $\pm$ 0.092	0.536 $\pm$ 0.100	0.524 $\pm$ 0.065	0.048 $\pm$ 0.028	0.053 $\pm$ 0.026
<i>NPLM-bilinear</i>								
plain	0.672 $\pm$ 0.089	0.805 $\pm$ 0.051	0.538 $\pm$ 0.095	0.544 $\pm$ 0.097	0.498 $\pm$ 0.092	0.504 $\pm$ 0.100	0.064 $\pm$ 0.067	0.070 $\pm$ 0.081

Table 41: Galaxy10 / VISReg: every (objective, construction) $\times$ (pre, post), across holdout draws. (B) Dataset-level power at $f = 0.02$, $\alpha = 0.05$. *Coverage:* Galaxy10 / VISReg, single holdout, 5 draws (classes 2,3,4,5,6), mean $\pm$ sd across draws, one seed per draw; plain-arm post columns = discovery from a 2% injected sample (per-fraction post geometry present in 5/5 draws, ‘–’ otherwise; the whole-class natural pass is in Table 42); residual concats from exp 71 (5 draws), post-discovery on the SupCon+res concat only (exp 72, 5 draws); concat SparKer pre from the archived draw (exp 80).

space	SparKer		Mahalanobis		MMD	
	pre	post	pre	post	pre	post
<i>SupCon</i>						
plain	0.10 $\pm$ 0.08	0.11 $\pm$ 0.03	0.10 $\pm$ 0.11	0.13 $\pm$ 0.10	0.20 $\pm$ 0.07	0.16 $\pm$ 0.12
+res concat	0.18	0.10 $\pm$ 0.11	–	0.10 $\pm$ 0.07	–	0.16 $\pm$ 0.13
+res-nplm concat	0.46	–	–	–	–	–
<i>SupCon+SIGReg (λ_5)</i>						
plain	0.10 $\pm$ 0.10	0.12 $\pm$ 0.08	0.12 $\pm$ 0.14	0.19 $\pm$ 0.20	0.16 $\pm$ 0.06	0.14 $\pm$ 0.08
+res concat	0.08	–	–	–	–	–
+res-nplm concat	–	–	–	–	–	–
<i>NPLM-dist. (sup.)</i>						
plain	0.18 $\pm$ 0.13	0.12 $\pm$ 0.05	0.16 $\pm$ 0.18	0.19 $\pm$ 0.09	0.10 $\pm$ 0.06	0.09 $\pm$ 0.04
<i>SimCLR</i>						
plain	0.02 $\pm$ 0.02	0.09 $\pm$ 0.08	0.11 $\pm$ 0.04	0.16 $\pm$ 0.15	0.03 $\pm$ 0.03	0.06 $\pm$ 0.04
<i>SimCLR+SIGReg</i>						
plain	0.06 $\pm$ 0.03	0.02 $\pm$ 0.02	0.04 $\pm$ 0.04	0.15 $\pm$ 0.14	0.10 $\pm$ 0.05	0.14 $\pm$ 0.10
<i>NPLM-bilinear</i>						
plain	0.07 $\pm$ 0.08	0.06 $\pm$ 0.08	0.06 $\pm$ 0.06	0.09 $\pm$ 0.05	0.08 $\pm$ 0.03	0.13 $\pm$ 0.15

C.4 galaxy10: supporting tables

Table 42: Galaxy10: the single-holdout battery across holdout draws, all three backbones. Pre-discovery probe, geometry and per-event power, then the fine-tuning loop’s post-discovery probe, tied Mahalanobis, round-1 pool purity (mean and worst draw) for *discovery from a 2% injected sample* of the held-out class (the paper’s regime). The last two columns are the natural pass, in which the whole held-out class is present in the unlabelled bank — a different, easier regime, shown for reference only. *Coverage:* Galaxy10, single holdout, draws [0, 3, 5, 7, 8] (distinct held-out classes), 18 (arm, base) rows; mean $\pm$ sd across draws, one seed per draw; injected-sample probe/mahaT present for 18/18 rows (‘-’ = npz predates the per-fraction keys); purity read from the run logs.

objective	pre-discovery					post, 2% injected				post, whole class	
	probe	acc	eucl	mahaT	per-ev	probe	mahaT	purity	min	probe	purity
<i>Galaxy10 / DINO, 5 draws</i>											
SupCon	0.879 $\pm$ 0.026	0.616 $\pm$ 0.013	0.526 $\pm$ 0.145	0.500 $\pm$ 0.135	0.063 $\pm$ 0.092	0.860 $\pm$ 0.032	0.488 $\pm$ 0.141	0.017 $\pm$ 0.021	0.000	0.864 $\pm$ 0.029	0.103 $\pm$ 0.138
SupCon+SIGReg ($\lambda=5$)	0.804 $\pm$ 0.064	0.588 $\pm$ 0.015	0.499 $\pm$ 0.104	0.482 $\pm$ 0.114	0.040 $\pm$ 0.038	0.849 $\pm$ 0.033	0.478 $\pm$ 0.097	0.017 $\pm$ 0.013	0.000	0.852 $\pm$ 0.042	0.092 $\pm$ 0.074
NPLM-dist. (sup.)	0.775 $\pm$ 0.097	0.442 $\pm$ 0.024	0.585 $\pm$ 0.123	0.556 $\pm$ 0.129	0.105 $\pm$ 0.121	0.782 $\pm$ 0.086	0.565 $\pm$ 0.092	0.043 $\pm$ 0.046	0.000	0.758 $\pm$ 0.132	0.131 $\pm$ 0.038
SimCLR	0.790 $\pm$ 0.064	0.449 $\pm$ 0.007	0.460 $\pm$ 0.125	0.476 $\pm$ 0.111	0.049 $\pm$ 0.046	0.810 $\pm$ 0.087	0.481 $\pm$ 0.111	0.012 $\pm$ 0.013	0.000	0.837 $\pm$ 0.057	0.104 $\pm$ 0.089
SimCLR+SIGReg	0.801 $\pm$ 0.084	0.440 $\pm$ 0.008	0.477 $\pm$ 0.069	0.508 $\pm$ 0.093	0.045 $\pm$ 0.029	0.846 $\pm$ 0.036	0.496 $\pm$ 0.070	0.014 $\pm$ 0.014	0.000	0.844 $\pm$ 0.042	0.100 $\pm$ 0.088
NPLM-bilinear	0.684 $\pm$ 0.134	0.380 $\pm$ 0.035	0.483 $\pm$ 0.063	0.450 $\pm$ 0.120	0.045 $\pm$ 0.036	0.811 $\pm$ 0.032	0.462 $\pm$ 0.125	0.010 $\pm$ 0.009	0.000	0.808 $\pm$ 0.038	0.104 $\pm$ 0.080
<i>Galaxy10 / LeJEPA, 5 draws</i>											
SupCon	0.892 $\pm$ 0.056	0.753 $\pm$ 0.018	0.591 $\pm$ 0.080	0.711 $\pm$ 0.115	0.047 $\pm$ 0.027	0.912 $\pm$ 0.053	0.713 $\pm$ 0.127	0.034 $\pm$ 0.008	0.025	0.920 $\pm$ 0.051	0.132 $\pm$ 0.085
SupCon+SIGReg ($\lambda=5$)	0.762 $\pm$ 0.181	0.696 $\pm$ 0.030	0.633 $\pm$ 0.148	0.665 $\pm$ 0.163	0.134 $\pm$ 0.119	0.827 $\pm$ 0.118	0.670 $\pm$ 0.124	0.043 $\pm$ 0.046	0.000	0.876 $\pm$ 0.064	0.182 $\pm$ 0.200
NPLM-dist. (sup.)	0.787 $\pm$ 0.073	0.627 $\pm$ 0.053	0.591 $\pm$ 0.133	0.613 $\pm$ 0.174	0.132 $\pm$ 0.095	0.798 $\pm$ 0.125	0.625 $\pm$ 0.130	0.068 $\pm$ 0.059	0.000	0.815 $\pm$ 0.107	0.190 $\pm$ 0.137
SimCLR	0.829 $\pm$ 0.066	0.521 $\pm$ 0.009	0.488 $\pm$ 0.086	0.574 $\pm$ 0.107	0.050 $\pm$ 0.039	0.847 $\pm$ 0.067	0.614 $\pm$ 0.047	0.012 $\pm$ 0.016	0.000	0.830 $\pm$ 0.101	0.112 $\pm$ 0.100
SimCLR+SIGReg	0.836 $\pm$ 0.060	0.460 $\pm$ 0.015	0.519 $\pm$ 0.134	0.596 $\pm$ 0.118	0.065 $\pm$ 0.070	0.846 $\pm$ 0.055	0.563 $\pm$ 0.101	0.037 $\pm$ 0.047	0.000	0.863 $\pm$ 0.049	0.123 $\pm$ 0.168
NPLM-bilinear	0.808 $\pm$ 0.091	0.533 $\pm$ 0.024	0.581 $\pm$ 0.065	0.611 $\pm$ 0.097	0.075 $\pm$ 0.028	0.835 $\pm$ 0.079	0.598 $\pm$ 0.087	0.038 $\pm$ 0.020	0.000	0.828 $\pm$ 0.101	0.102 $\pm$ 0.051
<i>Galaxy10 / VISReg, 5 draws</i>											
SupCon	0.906 $\pm$ 0.043	0.775 $\pm$ 0.019	0.700 $\pm$ 0.076	0.686 $\pm$ 0.074	0.095 $\pm$ 0.054	0.905 $\pm$ 0.041	0.672 $\pm$ 0.089	0.091 $\pm$ 0.044	0.048	0.907 $\pm$ 0.039	0.290 $\pm$ 0.067
SupCon+SIGReg ($\lambda=5$)	0.823 $\pm$ 0.108	0.744 $\pm$ 0.017	0.615 $\pm$ 0.118	0.581 $\pm$ 0.136	0.090 $\pm$ 0.096	0.907 $\pm$ 0.033	0.591 $\pm$ 0.130	0.053 $\pm$ 0.052	0.013	0.890 $\pm$ 0.060	0.146 $\pm$ 0.042
NPLM-dist. (sup.)	0.738 $\pm$ 0.105	0.616 $\pm$ 0.042	0.640 $\pm$ 0.112	0.628 $\pm$ 0.138	0.129 $\pm$ 0.098	0.733 $\pm$ 0.186	0.648 $\pm$ 0.112	0.089 $\pm$ 0.062	0.013	0.855 $\pm$ 0.056	0.262 $\pm$ 0.106
SimCLR	0.847 $\pm$ 0.053	0.564 $\pm$ 0.021	0.486 $\pm$ 0.076	0.526 $\pm$ 0.065	0.045 $\pm$ 0.019	0.838 $\pm$ 0.081	0.546 $\pm$ 0.062	0.005 $\pm$ 0.006	0.000	0.859 $\pm$ 0.051	0.093 $\pm$ 0.063
SimCLR+SIGReg	0.835 $\pm$ 0.040	0.437 $\pm$ 0.014	0.515 $\pm$ 0.084	0.536 $\pm$ 0.100	0.048 $\pm$ 0.028	0.861 $\pm$ 0.033	0.524 $\pm$ 0.065	0.012 $\pm$ 0.008	0.000	0.861 $\pm$ 0.027	0.110 $\pm$ 0.077
NPLM-bilinear	0.672 $\pm$ 0.089	0.360 $\pm$ 0.029	0.538 $\pm$ 0.095	0.498 $\pm$ 0.092	0.064 $\pm$ 0.067	0.805 $\pm$ 0.051	0.504 $\pm$ 0.100	0.022 $\pm$ 0.020	0.000	0.816 $\pm$ 0.047	0.117 $\pm$ 0.092

Table 43: Galaxy10: residual constructions across holdout draws. The paired probe gain of the concatenation over its own parent is the one Galaxy10 effect that survives draw variance. *Coverage:* Galaxy10, 3 backbones, single holdout, draws [0, 3, 5, 7, 8]; mean $\pm$ sd across draws; last column = paired concat—parent probe gain and wins/draws.

space	probe	acc	eucl	mahaT	per-ev	concat—parent (wins)
<i>Galaxy10 / DINO, 5 draws</i>						
supcon-ft (parent)	0.879 $\pm$ 0.026	0.616 $\pm$ 0.013	0.526 $\pm$ 0.145	0.500 $\pm$ 0.135	0.063 $\pm$ 0.092	—
supcon-ft-res (residual)	0.891 $\pm$ 0.025	0.526 $\pm$ 0.011	0.478 $\pm$ 0.107	0.467 $\pm$ 0.139	0.055 $\pm$ 0.048	—
supcon-ft-res (concat)	0.903 $\pm$ 0.027	0.594 $\pm$ 0.010	0.497 $\pm$ 0.122	0.488 $\pm$ 0.138	0.059 $\pm$ 0.078	+0.023 (5/5)
supcon-ft-res-nplm (residual)	0.862 $\pm$ 0.027	0.598 $\pm$ 0.011	0.553 $\pm$ 0.115	0.504 $\pm$ 0.111	0.067 $\pm$ 0.066	—
supcon-ft-res-nplm (concat)	0.890 $\pm$ 0.023	0.619 $\pm$ 0.012	0.538 $\pm$ 0.140	0.494 $\pm$ 0.134	0.062 $\pm$ 0.088	+0.011 (5/5)
ss-ft (parent)	0.804 $\pm$ 0.064	0.588 $\pm$ 0.015	0.499 $\pm$ 0.104	0.482 $\pm$ 0.114	0.040 $\pm$ 0.038	—
ss-ft-res (residual)	0.852 $\pm$ 0.033	0.522 $\pm$ 0.014	0.493 $\pm$ 0.084	0.487 $\pm$ 0.101	0.056 $\pm$ 0.042	—
ss-ft-res (concat)	0.869 $\pm$ 0.026	0.568 $\pm$ 0.013	0.494 $\pm$ 0.089	0.486 $\pm$ 0.116	0.052 $\pm$ 0.040	+0.065 (5/5)
<i>Galaxy10 / LeJEPA, 5 draws</i>						
supcon-ft (parent)	0.892 $\pm$ 0.056	0.753 $\pm$ 0.018	0.591 $\pm$ 0.080	0.711 $\pm$ 0.115	0.047 $\pm$ 0.027	—
supcon-ft-res (residual)	0.910 $\pm$ 0.045	0.638 $\pm$ 0.018	0.623 $\pm$ 0.058	0.643 $\pm$ 0.090	0.114 $\pm$ 0.057	—
supcon-ft-res (concat)	0.945 $\pm$ 0.033	0.668 $\pm$ 0.016	0.630 $\pm$ 0.059	0.709 $\pm$ 0.079	0.124 $\pm$ 0.059	+0.053 (5/5)
supcon-ft-res-nplm (residual)	0.887 $\pm$ 0.049	0.702 $\pm$ 0.025	0.596 $\pm$ 0.119	0.598 $\pm$ 0.105	0.083 $\pm$ 0.066	—
supcon-ft-res-nplm (concat)	0.931 $\pm$ 0.045	0.760 $\pm$ 0.022	0.644 $\pm$ 0.080	0.699 $\pm$ 0.089	0.086 $\pm$ 0.061	+0.039 (5/5)
ss-ft (parent)	0.762 $\pm$ 0.181	0.696 $\pm$ 0.030	0.633 $\pm$ 0.148	0.665 $\pm$ 0.163	0.134 $\pm$ 0.119	—
ss-ft-res (residual)	0.854 $\pm$ 0.075	0.611 $\pm$ 0.010	0.521 $\pm$ 0.123	0.589 $\pm$ 0.132	0.061 $\pm$ 0.060	—
ss-ft-res (concat)	0.855 $\pm$ 0.094	0.706 $\pm$ 0.020	0.627 $\pm$ 0.068	0.692 $\pm$ 0.118	0.093 $\pm$ 0.047	+0.093 (5/5)
<i>Galaxy10 / VISReg, 5 draws</i>						
supcon-ft (parent)	0.906 $\pm$ 0.043	0.775 $\pm$ 0.019	0.700 $\pm$ 0.076	0.686 $\pm$ 0.074	0.095 $\pm$ 0.054	—
supcon-ft-res (residual)	0.930 $\pm$ 0.036	0.691 $\pm$ 0.030	0.528 $\pm$ 0.068	0.606 $\pm$ 0.052	0.056 $\pm$ 0.055	—
supcon-ft-res (concat)	0.947 $\pm$ 0.025	0.769 $\pm$ 0.020	0.647 $\pm$ 0.065	0.658 $\pm$ 0.061	0.087 $\pm$ 0.054	+0.041 (5/5)
supcon-ft-res-nplm (residual)	0.890 $\pm$ 0.053	0.774 $\pm$ 0.017	0.685 $\pm$ 0.109	0.709 $\pm$ 0.109	0.114 $\pm$ 0.101	—
supcon-ft-res-nplm (concat)	0.923 $\pm$ 0.036	0.781 $\pm$ 0.019	0.709 $\pm$ 0.097	0.711 $\pm$ 0.106	0.116 $\pm$ 0.088	+0.017 (5/5)
ss-ft (parent)	0.823 $\pm$ 0.108	0.744 $\pm$ 0.017	0.615 $\pm$ 0.118	0.581 $\pm$ 0.136	0.090 $\pm$ 0.096	—
ss-ft-res (residual)	0.867 $\pm$ 0.054	0.645 $\pm$ 0.023	0.498 $\pm$ 0.082	0.554 $\pm$ 0.102	0.048 $\pm$ 0.032	—
ss-ft-res (concat)	0.888 $\pm$ 0.052	0.740 $\pm$ 0.018	0.576 $\pm$ 0.095	0.586 $\pm$ 0.136	0.086 $\pm$ 0.062	+0.065 (5/5)

Table 44: Supervised top-1 (exp 132) on Galaxy10: archived draw and the draw interval. Sup-Con+SIGReg costs 0.035 against SupCon on every draw; the residual concats tie the parent. *Coverage:* 13 cell files; 3 probe restarts per cell; draws are the exp-125 single-holdout draws.

space	top-1 (archived)	top-1 (draws)	n
<i>Galaxy10 / DINO</i>			
supcon-ft	0.642	0.666 ± 0.011	5
ss-ft	0.597	0.628 ± 0.013	5
nplm-sup-ft	0.474	0.525 ± 0.023	5
simclr-ft	0.556	0.584 ± 0.010	5
sigreg-ssl-ft	0.533	0.562 ± 0.011	5
nplm-bil-ft	0.404	0.537 ± 0.028	5
supcon-ft-res-cat	0.641	0.663 ± 0.010	2
supcon-ft-resnplm-cat	0.642	0.654	1
ss-ft-res-cat	0.599	0.624 ± 0.009	2
<i>Galaxy10 / LeJEPa</i>			
supcon-ft	0.781	0.790 ± 0.018	5
ss-ft	0.745	0.755 ± 0.024	5
nplm-sup-ft	0.704	0.710 ± 0.029	5
simclr-ft	0.617	0.631 ± 0.018	5
sigreg-ssl-ft	0.543	0.582 ± 0.014	5
nplm-bil-ft	0.579	0.614 ± 0.018	5
supcon-ft-res-cat	0.775	—	0
supcon-ft-resnplm-cat	0.775	—	0
ss-ft-res-cat	0.743	—	0
<i>Galaxy10 / VISReg</i>			
supcon-ft	0.776	—	0
ss-ft	0.757	—	0
nplm-sup-ft	0.636	—	0
simclr-ft	0.612	—	0
sigreg-ssl-ft	0.506	—	0
nplm-bil-ft	0.480	—	0
supcon-ft-res-cat	0.777	—	0
supcon-ft-resnplm-cat	0.777	—	0
ss-ft-res-cat	0.756	—	0

D DTD: every measurement, across holdout draws

DTD is task-novel to the backbones (textures) and is the dataset on which the default discovery objective was chosen. Single holdout is one texture class of 47 ($b \approx 0.02$), resampled over classes 8, 46, 38, 15 and 16; Table 45 gives all three backbones. The multi-holdout regime (10 classes held out, $b \approx 0.2$) is the one the earlier report quoted, and its purities (0.80) are not comparable with these (0.22).

Table 45: DTD: the single-holdout battery across holdout draws, all three backbones. Pre-discovery probe, geometry and per-event power, then the fine-tuning loop’s post-discovery probe, tied Mahalanobis, round-1 pool purity (mean and worst draw) for *discovery from a 2% injected sample* of the held-out class (the paper’s regime). The last two columns are the natural pass, in which the whole held-out class is present in the unlabelled bank — a different, easier regime, shown for reference only. *Coverage:* DTD, single holdout, draws [0, 1, 3, 4, 5] (distinct held-out classes), 18 (arm, base) rows; mean $\pm$ sd across draws, one seed per draw; injected-sample probe/mahaT present for 18/18 rows (‘-’ = npz predates the per-fraction keys); purity read from the run logs.

objective	pre-discovery					post, 2% injected				post, whole class	
	probe	acc	eucl	mahaT	per-ev	probe	mahaT	purity	min	probe	purity
<i>DTD / DINO, 5 draws</i>											
SupCon	0.883 $\pm$ 0.040	0.739 $\pm$ 0.004	0.732 $\pm$ 0.045	0.628 $\pm$ 0.077	0.020 $\pm$ 0.010	0.908 $\pm$ 0.040	0.689 $\pm$ 0.054	0.028 $\pm$ 0.012	0.011	0.892 $\pm$ 0.070	0.028 $\pm$ 0.012
SupCon+SIGReg ($\lambda=5$)	0.787 $\pm$ 0.036	0.762 $\pm$ 0.005	0.792 $\pm$ 0.019	0.732 $\pm$ 0.054	0.230 $\pm$ 0.053	0.645 $\pm$ 0.094	0.690 $\pm$ 0.031	0.211 $\pm$ 0.015	0.186	0.702 $\pm$ 0.056	0.219 $\pm$ 0.014
NPLM-dist. (sup.)	0.749 $\pm$ 0.102	0.445 $\pm$ 0.018	0.632 $\pm$ 0.066	0.635 $\pm$ 0.079	0.055 $\pm$ 0.043	0.828 $\pm$ 0.085	0.634 $\pm$ 0.129	0.049 $\pm$ 0.026	0.021	0.861 $\pm$ 0.044	0.051 $\pm$ 0.027
SimCLR	0.940 $\pm$ 0.039	0.631 $\pm$ 0.004	0.550 $\pm$ 0.100	0.564 $\pm$ 0.111	0.040 $\pm$ 0.080	0.885 $\pm$ 0.043	0.560 $\pm$ 0.124	0.004 $\pm$ 0.008	0.000	0.906 $\pm$ 0.040	0.005 $\pm$ 0.010
SimCLR+SIGReg	0.890 $\pm$ 0.070	0.624 $\pm$ 0.010	0.627 $\pm$ 0.103	0.606 $\pm$ 0.090	0.050 $\pm$ 0.042	0.820 $\pm$ 0.096	0.591 $\pm$ 0.100	0.003 $\pm$ 0.002	0.000	0.834 $\pm$ 0.076	0.003 $\pm$ 0.002
NPLM-bilinear	0.728 $\pm$ 0.113	0.340 $\pm$ 0.073	0.513 $\pm$ 0.075	0.502 $\pm$ 0.081	0.020 $\pm$ 0.019	0.853 $\pm$ 0.051	0.587 $\pm$ 0.077	0.004 $\pm$ 0.005	0.000	0.879 $\pm$ 0.040	0.004 $\pm$ 0.005
<i>DTD / LeJEPa, 5 draws</i>											
SupCon	0.659 $\pm$ 0.144	0.728 $\pm$ 0.004	0.722 $\pm$ 0.070	0.712 $\pm$ 0.084	0.110 $\pm$ 0.162	0.738 $\pm$ 0.109	0.752 $\pm$ 0.116	0.068 $\pm$ 0.071	0.011	0.758 $\pm$ 0.069	0.073 $\pm$ 0.077
SupCon+SIGReg ($\lambda=5$)	0.803 $\pm$ 0.059	0.763 $\pm$ 0.005	0.787 $\pm$ 0.024	0.780 $\pm$ 0.027	0.245 $\pm$ 0.073	0.625 $\pm$ 0.159	0.758 $\pm$ 0.039	0.216 $\pm$ 0.004	0.210	0.692 $\pm$ 0.123	0.225 $\pm$ 0.006
NPLM-dist. (sup.)	0.708 $\pm$ 0.097	0.614 $\pm$ 0.010	0.811 $\pm$ 0.056	0.804 $\pm$ 0.035	0.195 $\pm$ 0.093	0.677 $\pm$ 0.089	0.789 $\pm$ 0.065	0.208 $\pm$ 0.025	0.182	0.648 $\pm$ 0.158	0.215 $\pm$ 0.024
SimCLR	0.938 $\pm$ 0.037	0.618 $\pm$ 0.006	0.563 $\pm$ 0.111	0.588 $\pm$ 0.113	0.030 $\pm$ 0.048	0.897 $\pm$ 0.056	0.706 $\pm$ 0.131	0.001 $\pm$ 0.002	0.000	0.912 $\pm$ 0.045	0.002 $\pm$ 0.002
SimCLR+SIGReg	0.909 $\pm$ 0.057	0.605 $\pm$ 0.006	0.656 $\pm$ 0.072	0.674 $\pm$ 0.108	0.085 $\pm$ 0.066	0.839 $\pm$ 0.067	0.701 $\pm$ 0.089	0.020 $\pm$ 0.019	0.000	0.888 $\pm$ 0.039	0.021 $\pm$ 0.019
NPLM-bilinear	0.798 $\pm$ 0.077	0.463 $\pm$ 0.071	0.583 $\pm$ 0.050	0.648 $\pm$ 0.034	0.055 $\pm$ 0.010	0.890 $\pm$ 0.062	0.664 $\pm$ 0.033	0.029 $\pm$ 0.014	0.016	0.873 $\pm$ 0.063	0.033 $\pm$ 0.016
<i>DTD / VISReg, 5 draws</i>											
SupCon	0.774 $\pm$ 0.101	0.762 $\pm$ 0.005	0.781 $\pm$ 0.052	0.730 $\pm$ 0.088	0.175 $\pm$ 0.142	0.808 $\pm$ 0.102	0.736 $\pm$ 0.082	0.172 $\pm$ 0.031	0.124	0.781 $\pm$ 0.100	0.184 $\pm$ 0.031
SupCon+SIGReg ($\lambda=5$)	0.867 $\pm$ 0.028	0.788 $\pm$ 0.007	0.802 $\pm$ 0.049	0.782 $\pm$ 0.054	0.285 $\pm$ 0.066	0.739 $\pm$ 0.129	0.780 $\pm$ 0.046	0.229 $\pm$ 0.023	0.189	0.737 $\pm$ 0.114	0.241 $\pm$ 0.025
NPLM-dist. (sup.)	0.702 $\pm$ 0.121	0.584 $\pm$ 0.010	0.747 $\pm$ 0.055	0.757 $\pm$ 0.034	0.130 $\pm$ 0.048	0.609 $\pm$ 0.127	0.802 $\pm$ 0.036	0.154 $\pm$ 0.051	0.061	0.656 $\pm$ 0.083	0.163 $\pm$ 0.054
SimCLR	0.899 $\pm$ 0.088	0.606 $\pm$ 0.003	0.515 $\pm$ 0.135	0.523 $\pm$ 0.122	0.070 $\pm$ 0.073	0.903 $\pm$ 0.053	0.593 $\pm$ 0.126	0.021 $\pm$ 0.028	0.000	0.904 $\pm$ 0.077	0.023 $\pm$ 0.032
SimCLR+SIGReg	0.923 $\pm$ 0.053	0.610 $\pm$ 0.004	0.664 $\pm$ 0.117	0.664 $\pm$ 0.129	0.130 $\pm$ 0.087	0.919 $\pm$ 0.044	0.659 $\pm$ 0.125	0.021 $\pm$ 0.022	0.000	0.911 $\pm$ 0.049	0.023 $\pm$ 0.024
NPLM-bilinear	0.791 $\pm$ 0.093	0.458 $\pm$ 0.103	0.567 $\pm$ 0.111	0.576 $\pm$ 0.153	0.040 $\pm$ 0.025	0.862 $\pm$ 0.040	0.590 $\pm$ 0.167	0.012 $\pm$ 0.006	0.005	0.854 $\pm$ 0.052	0.012 $\pm$ 0.006

E Objective naming

Three objectives collide under similar names at two regulariser strengths, and their numbers must not be pooled.

paper name	code name(s)	objective	λ
SimCLR+SIGReg	<code>ss</code> , <code>simclr_sigreg</code>	instance / cosine / softmax, unsupervised	1
SupCon+SIGReg (weak)	<code>supcon_sigreg</code>	supervised / cosine / softmax, $\tau=0.1$	1
SupCon+SIGReg (strong)	<code>ss-ft</code> , <code>ssig</code>	same objective as the row above	5
SupCon+SIGReg (repulse)	<code>supsig</code>	the proto/repulse recipe; <i>not</i> SupCon+SIGReg	–
NPLM-dist. + classwise SIGReg	<code>nplmcw</code> , <code>nplm_dist_sup_cw</code>	supervised / distance / NPLM, class-wise marginal	5
NPLM-dist. (sup.)	<code>nplmsd</code> , <code>nplm-sup-ft</code>	supervised / distance / NPLM, global marginal	1
SupCon+NPLM concat	<code>supcon+nplm</code>	concatenation of a 16-D SupCon head and a 16-D NPLM head (hub lineage, exp 59)	–

The strong and weak SupCon+SIGReg variants are the same objective at different λ (aux-term magnitudes 0.381 vs. 0.066 on a shared batch); the paper default is the strong one, and every exp-50-era number is the weak one. `ss` and `ss-ft` share an abbreviation and are different algorithms, one unsupervised and one supervised. The from-scratch arms `ssig` and `nplmsd` compute numerically identical losses to their fine-tune twins `ss-ft` and `nplm-sup-ft`. A final source of collision: LeJEPa and VISReg name pretrained *backbones* throughout this paper; where a training objective of the same family appears (the unsupervised arms of the design cube) we write it as an objective name (SimCLR+SIGReg), never by the backbone’s name.